Data Science

Lecture 02: Probability Theory

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Where We Left Off

Lecture 01 ended on a claim: data does not speak for itself — an assumption always speaks for it, and a pattern is knowledge only once it has been checked. Today builds the language in which both are written.

Assume

Modules 1–5 — a model of chance: outcomes, probabilities, random variables, the joint

→

Observe

Module 6 — a sample, described, with its error

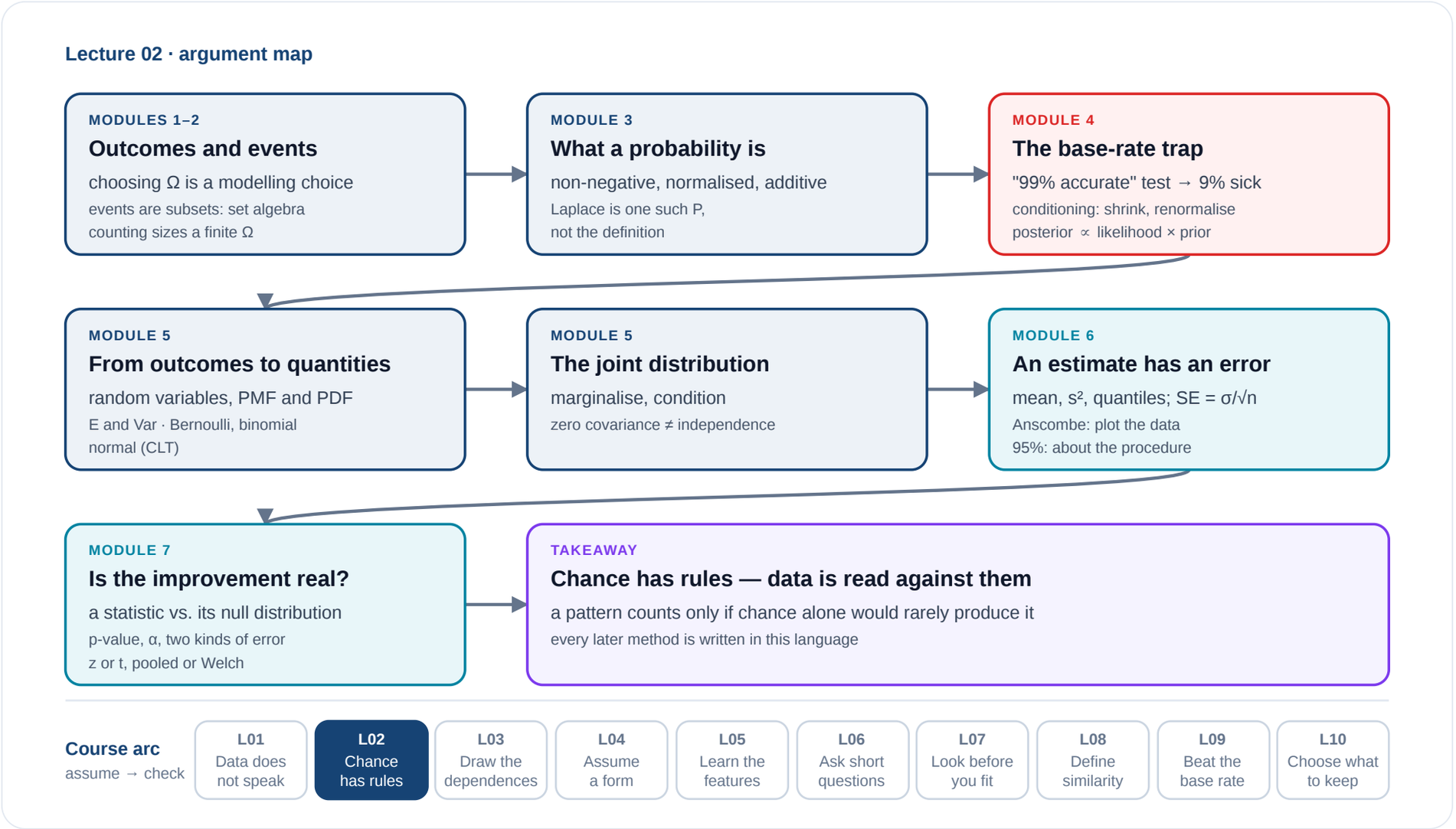
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Check

Module 7 — would chance alone produce this?

> The app users of Lecture 01 churn less. By the end of today you can say whether that difference is real — not yet *why* (that is Lecture 03).

Argument Map



Module 0

Orientation

> ***Guiding Question:*** *What does a data scientist actually need from probability theory — and why does the formal treatment come first?*

The Landscape

Probability theory
Given a model of chance, what follows?
Kolmogorov axioms, distributions, limit theorems

Mathematical statistics
Given data, which model of chance is plausible?
Estimation, hypothesis testing

Data science
Given data and a decision, what should we do?
Prediction, learning, evaluation

The arrow runs left to right: this lecture builds the vocabulary the rest of the course spends its time using.

What We Need From Probability

In this lecture

- sample spaces and events
- counting outcomes
- probability as a measure
- conditioning and independence
- Bayes' theorem
- random variables and distributions
- describing data numerically
- testing a hypothesis

Where it comes back

- naive Bayes and Bayesian decisions
- impurity measures in decision trees
- likelihoods in probabilistic models
- sampling and error estimates in evaluation
- independence assumptions in graphical models



Example: Two Fruit Boxes

Red box
2 apples, 6 oranges

Blue box
3 apples, 1 orange

We pick a box, then a fruit from it.

- If we pick the **red box 40%** of the time, how often do we get an apple?
- We got an **orange** — which box did it probably come from?

The Counts, in One Grid

A hundred picks with the red box chosen 40% of the time, as a table of counts — box across, fruit down:

	$B = r$	$B = b$	row total
$F = a$ (apple)	10	45	**55**
$F = o$ (orange)	30	15	**45**
column total	**40**	**60**	**100**

Joint
a single cell over the total
$P(B = r, F = a) = \frac{10}{100}$

Marginal
a row or column total over the total
$P(F = a) = \frac{55}{100}$

Conditional
a cell over its own row or column
$P(F = a \mid B = r) = \frac{10}{40}$

The Two Rules

Sum rule

$$P(X) = \sum_{Y} P(X, Y)$$

Add along a row or column to remove a variable.

Product rule

$$P(X, Y) = P(Y \mid X) P(X)$$

A cell is its column total times its share of that column.

Almost all of probability theory is these two rules applied repeatedly.

Bayes' Theorem, in Advance

Combining the two rules and the symmetry $P(X, Y) = P(Y, X)$:

$$P(Y \mid X) = \frac{P(X \mid Y)P(Y)}{P(X)}, \qquad P(X) = \sum_{Y} P(X \mid Y)P(Y)$$

Which box did the orange come from?

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Which box did the orange come from?

$$P(B = r \mid F = o) = \frac{P(F = o \mid B = r)P(B = r)}{P(F = o)} = \frac{\frac{6}{8} \cdot 0.4}{\frac{6}{8} \cdot 0.4 + \frac{1}{4} \cdot 0.6} = \frac{2}{3}$$

Before the pick: 40% red. After seeing an orange: 67% red. The orange is **evidence** — because the red box is full of oranges and the blue one is not.

Bayes' Theorem, in Advance

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Before the pick: 40% red. After seeing an orange: 67% red. The orange is **evidence** — because the red box is full of oranges and the blue one is not.

And if the blue box held 3 apples and 9 oranges — the same orange share as the red one?

Takeaway

> Two rules — **sum** to remove a variable, **product** to split a joint — generate joint, marginal, conditional probabilities and Bayes' theorem. What follows makes them precise and extends them beyond counting.

Module 1

Events and the Algebra of Sets

> ***Guiding Question:*** *If probability is going to be a measure, what exactly are the objects it measures — and how do they combine?*

What Makes an Experiment "Random"?

Definition (Random experiment). A procedure that can — at least in theory — be repeated infinitely often, characterised by:

1. **Configuration** — a precisely specified system that can be reconstructed.
2. **Procedure** — an instruction for executing the experiment.
3. **Unpredictability** of the outcome.

Coin flip: a 1-Euro coin (configuration), placed on the thumb and snapped into the air (procedure), record heads or tails (outcome).

Randomness as Missing Information

> A random experiment is still causal. Its unpredictability reflects **missing information about the causal chain**, not an absence of causes.
>
> Consequence: a random experiment can become a deterministic process once new information is available.

A coin flip filmed at 10,000 fps and measured precisely enough stops being random. What does that imply for a model that calls something "random"?

Sample Space and Events

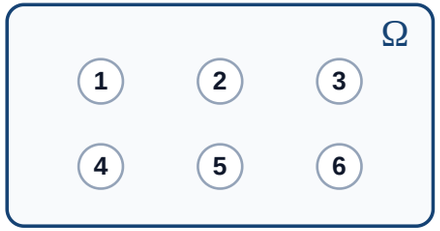
Definition (Sample space, event space). A set $\Omega = \{\omega_1, \omega_2, \ldots, \omega_n\}$ is the **sample space** of a random experiment if every outcome is associated with exactly one element $\omega \in \Omega$. The elements are called **outcomes**.

Let Ω be finite. Every subset $A \subseteq \Omega$ is an **event**; the event A **occurs** iff the observed outcome ω is a member of A. The set of all events, the power set $\mathcal{P}(\Omega)$, is the **event space**.

An outcome is a single result; an event is a *question you can ask* about the result.

For infinite Ω — the drug dosage below — the event space is a suitable family of subsets rather than the full power set; we return to this when continuous distributions appear.

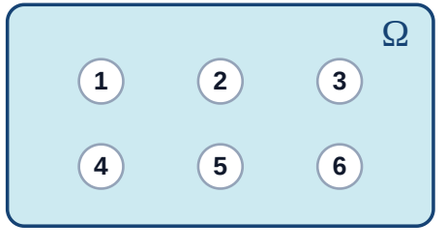
Important Event Types



$\varnothing$

impossible event

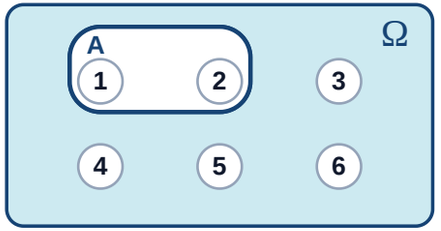
no outcome belongs to it



Ω

certain event

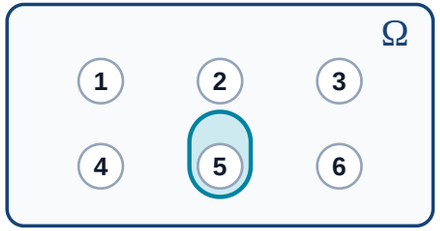
every outcome belongs to it



$\bar{A} = \Omega \setminus A$

complementary event

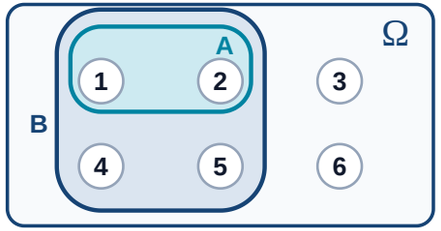
A = {1, 2}, $\bar{A}$ = {3, 4, 5, 6}



$|A| = 1$

elementary event

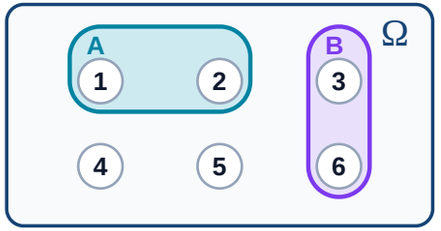
A = {5}: one outcome



$A \subseteq B$

A entails B

A = {1, 2}, B = {1, 2, 4, 5}



$A \cap B = \varnothing$

incompatible events

A = {1, 2}, B = {3, 6}

Examples

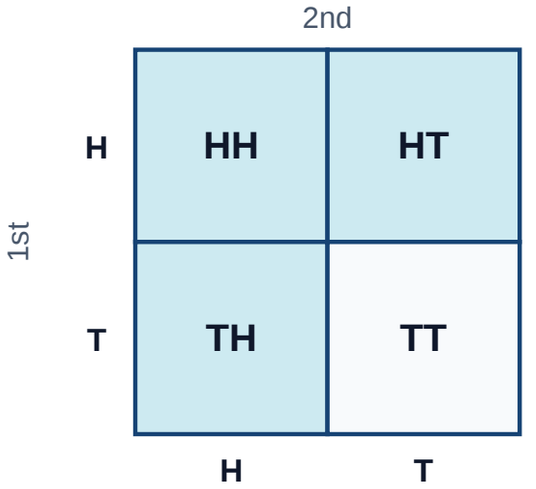
One coin flip



E = {H}: "heads"

all events: ∅, {H}, {T}, {H, T}

Two coin flips



"at least one head"

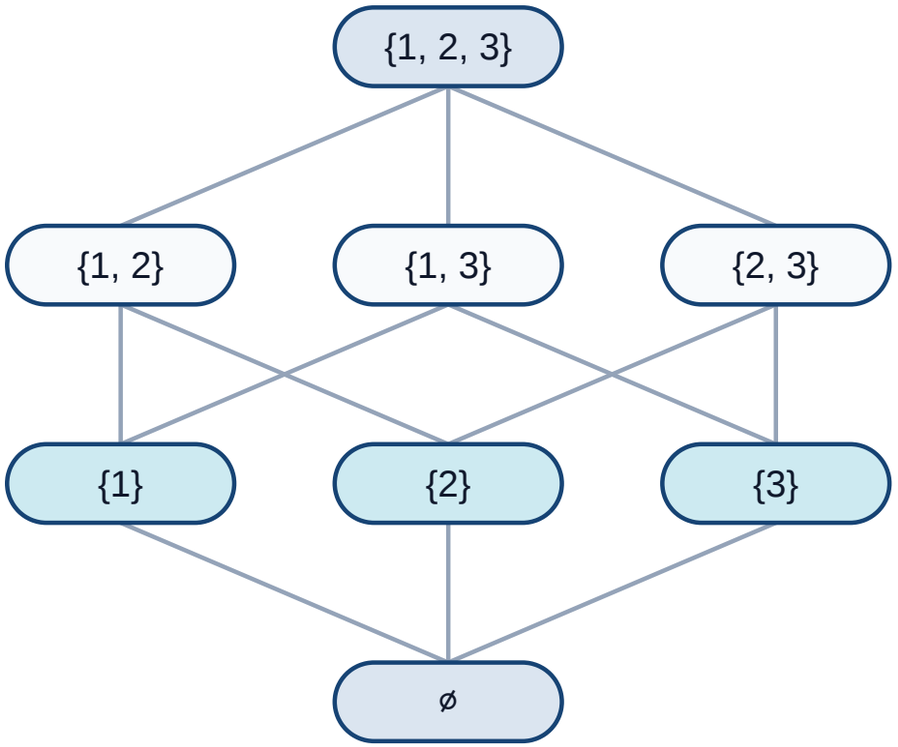
= {HH, HT, TH}

4 outcomes

2^4 = 16 possible events

An event is a set of outcomes you can shade — "heads", "at least one head" — and the sample space fixes which such questions can be asked.

The Event Space Grows Fast



$|\Omega| = n \rightarrow 2^n$ events

1	2
2	4
3	8
6	64
10	1,024
20	1,048,576

lines: one element added

For $|\Omega| = n$ the event space has 2^n elements — far too many to list, which is why probability is done by counting, not by enumeration.

Not Every Sample Space Is Finite

Dosage of a drug: Ω = (0, ∞)



uncountably many outcomes — probability can only live on intervals, not on points

Race with 7 teams: all orderings

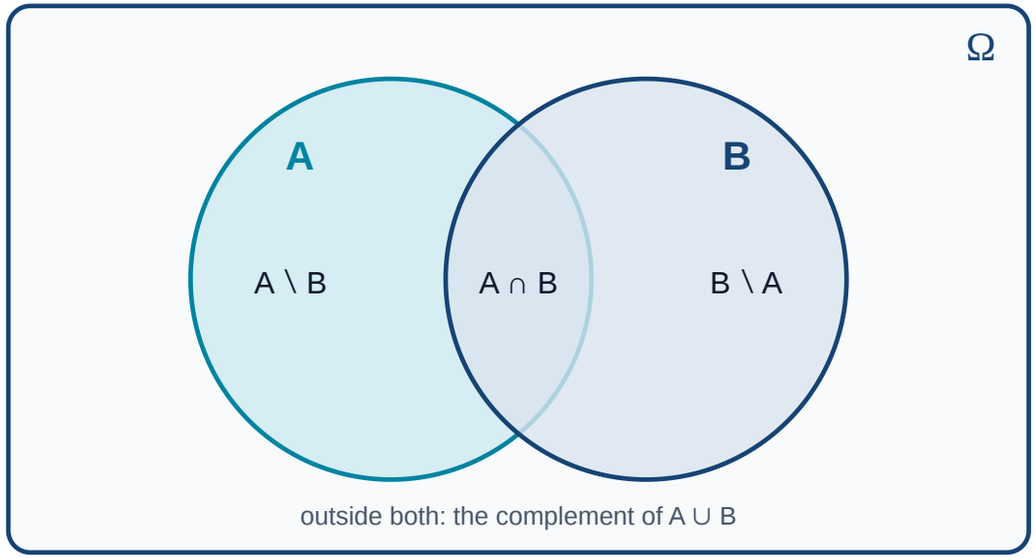


7! = 5,040 outcomes; shaded: the event "team 3 wins"

Choosing Ω is a modelling decision: it fixes what counts as an outcome, and therefore which questions can be asked at all.

Venn Diagrams

Venn diagrams are the graphical representation of events:

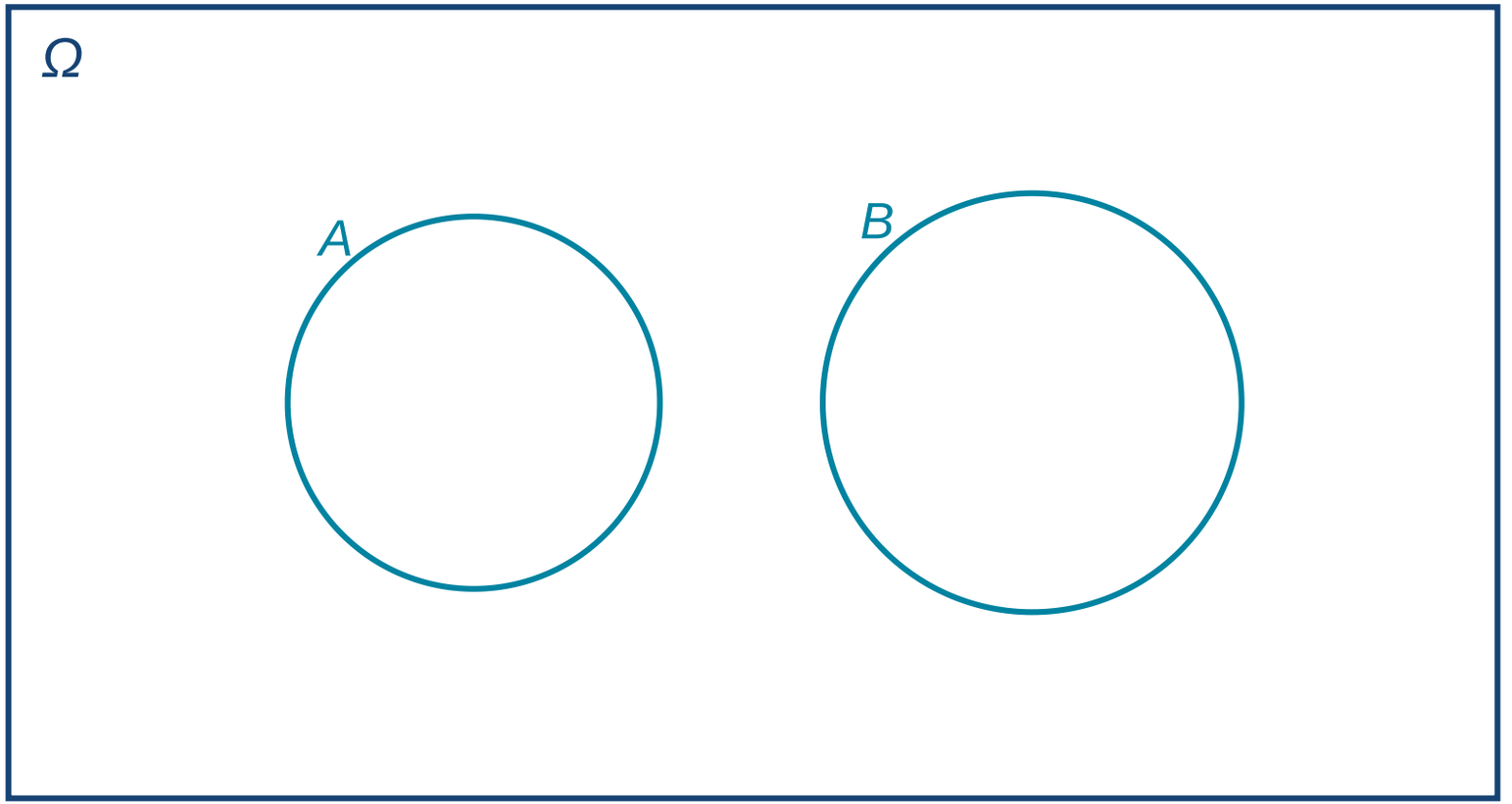


- the **rectangle** is the sample space Ω
- a **circle** is an event $A \subseteq \Omega$
- every **region** is a set — and every set operation is a region you can point at

Important Event Types

Before each step: which region will light up?

$$A \cap B = \emptyset$$



A and B are disjoint

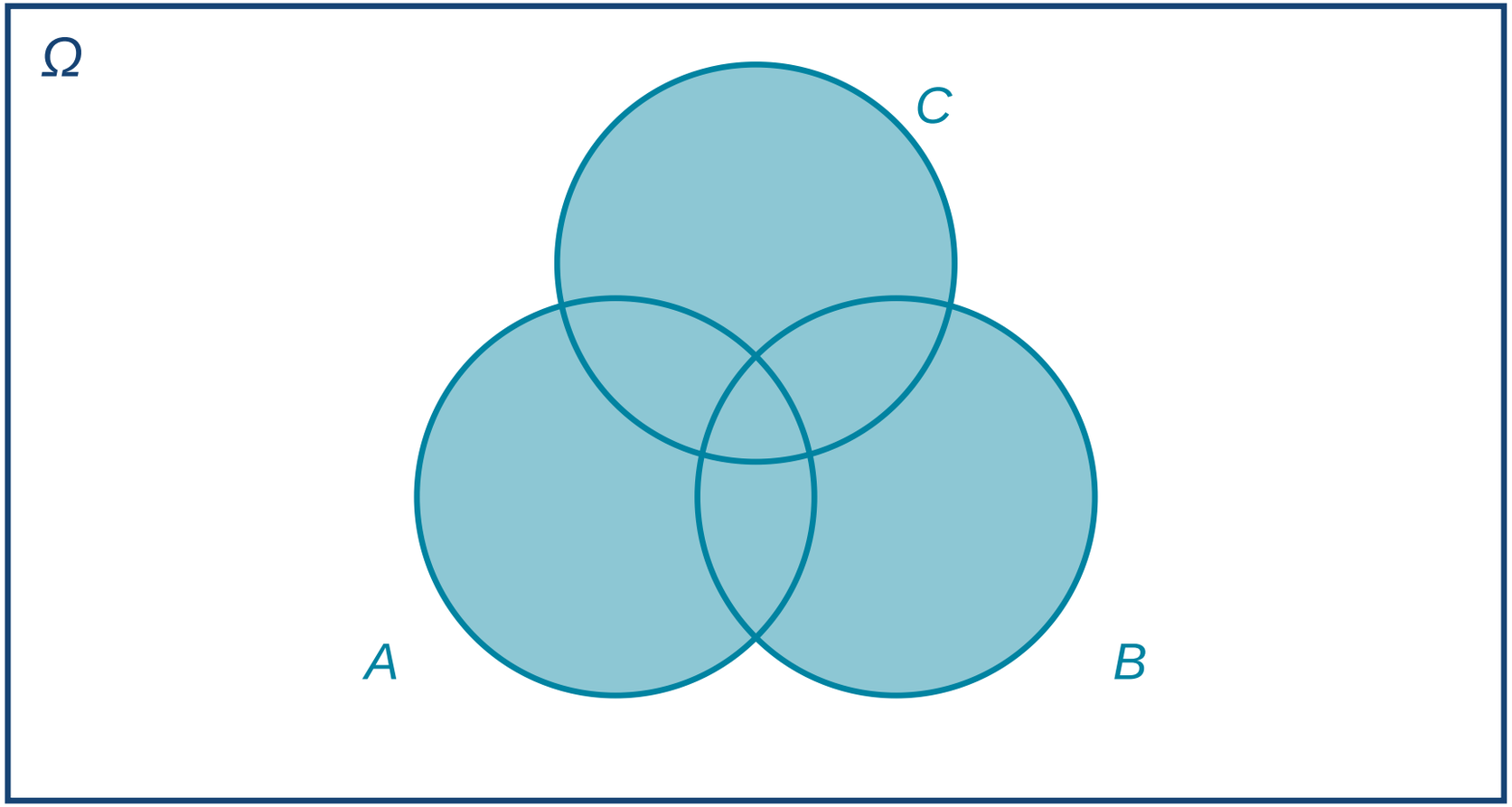
Algebra of Events

Law	Union form	Intersection form
Commutative	$A \cup B = B \cup A$	$A \cap B = B \cap A$
Associative	$(A \cup B) \cup C = A \cup (B \cup C)$	$(A \cap B) \cap C = A \cap (B \cap C)$
Distributive	$(A \cup B) \cap C = (A \cap C) \cup (B \cap C)$	$(A \cap B) \cup C = (A \cup C) \cap (B \cup C)$

Events form a set algebra — the same laws you know, but now every identity is a picture.

Set Operations as Regions

A ∪ B ∪ C



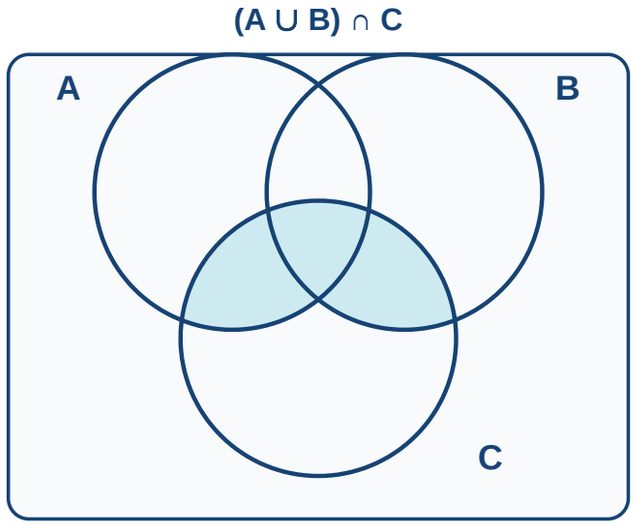
Union of three events

Mini-Exercise

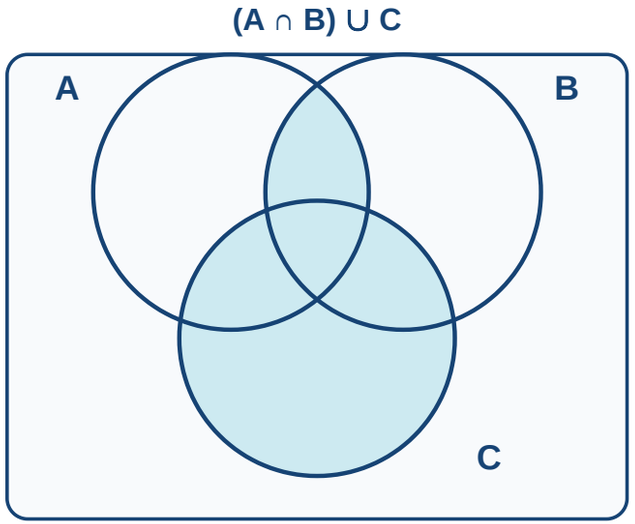
Are $(A \cup B) \cap C$ and $(A \cap B) \cup C$ the same region?

Mini-Exercise

Are $(A \cup B) \cap C$ and $(A \cap B) \cup C$ the same region?



inside C only



all of C, plus the lens A ∩ B

No. The first lies entirely inside C; the second is all of C plus the lens $A \cap B$. The distributive laws expand them into different expressions.

Takeaway

An event is a **subset** of the sample space, and combining statements about outcomes is set algebra: *or* is a union, *and* an intersection, *not* a complement. Choosing Ω decides which events exist at all.

Module 2

Counting

> ***Guiding Question:*** *How do we determine the size of a sample space we could never write down?*

Why Counting Comes First

If every outcome in a finite sample space is equally likely, then the probability of an event is a **ratio of counts**:

$$P(A) = \frac{|A|}{|\Omega|}$$

So before we can measure probability, we must be able to count $|A|$ and $|\Omega|$ — often without enumerating them.

Basic Principle of Counting

Generalised basic principle of counting. If an experiment consists of r stages, where stage i has n_i possible outcomes regardless of the earlier stages, then the whole experiment has

$$n_1 \cdot n_2 \cdots n_r$$

possible outcomes.

Example. How many 7-place licence plates, if the first 3 places are letters (26) and the last 4 are digits (10)?

$$26^3 \cdot 10^4 = 175{,}760{,}000$$

Three Questions Decide the Formula

Order?

Does the sequence matter?
matters → **permutation** (rankings)
does not → **combination** (committees)

Repetition?

May an element be picked twice?
without — teams of people
with — letters in a word

How many?

Select k of n elements, or arrange all n?

The Four Basic Cases

Ordered, without repetition

k-permutation of n:

$$\frac{n!}{(n-k)!}$$

From $\{a, b\}$, $k = 2$: ab, ba

Ordered, with repetition

$$n^k$$

From $\{a, b\}$, $k = 2$: aa, ab, ba, bb

Unordered, without repetition

k-combination of n:

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}$$

From $\{a, b\}$, $k = 2$: $\{a, b\}$

Unordered, with repetition

$$\binom{n+k-1}{k}$$

From $\{a, b\}$, $k = 2$: $\{a, a\}, \{a, b\}, \{b, b\}$

Permutations

Definition (k-permutation). An ordered selection of k elements from a set of n distinct elements without repetition. Counting position by position: n choices, then $n-1$, …, down to $n-k+1$, giving $\frac{n!}{(n-k)!}$.

Example. A class of 6 men and 4 women is ranked after an exam, no ties. How many possible rankings? $10! = 3{,}628{,}800$.

Combinations

Each k-combination corresponds to $k!$ different k-permutations — the same k elements in every possible order. Therefore

$$\binom{n}{k} = \frac{\text{number of } k\text{-permutations}}{k!} = \frac{n!}{k!(n-k)!}$$

Example. A committee of 3 from 20 people: $\binom{20}{3} = 1140$.

Dividing by $k!$ is the standard move for "forgetting" an ordering we counted.
Remember it as *over-count, then correct*.

Properties of the Binomial Coefficient

$$\binom{n}{k} = \binom{n}{n-k}$$

Choosing which k to include is the same as choosing which $n-k$ to leave out.

$$\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}$$

Split by a fixed element i: groups **containing** i, plus groups **not containing** i.

Binomial theorem.

$$(x+y)^n = \sum_{k=0}^{n} \binom{n}{k} x^k y^{n-k}$$

Application: How Many Subsets?

A set with n elements has

$$\sum_{k=0}^{n} \binom{n}{k} = 2^n$$

subsets — sum over all subset sizes.

> **Alternative intuition.** Each element is either in the subset or not: a binary choice, n times, so $2 \cdot 2 \cdots 2 = 2^n$.

This is the size of the event space $\mathcal{P}(\Omega)$ from the sample-space unit.

Mini-Exercise

From 5 people, how many ways are there to pick a **chair and a deputy**? And how many ways to pick a **two-person committee**?

Mini-Exercise

From 5 people, how many ways are there to pick a **chair and a deputy**? And how many ways to pick a **two-person committee**?

Chair and deputy: order matters, $5 \cdot 4 = 20$. Committee: each pair was counted twice (as ab and ba), so $20/2! = 10 = \binom{5}{2}$.

Takeaway

> Counting reduces to three questions — **order? repetition? how many?** — and one habit: when an ordering was counted that should not have been, divide it out.

Module 3

What Is a Probability?

> ***Guiding Question:*** *Every attempt to define probability turns out to be circular. What do we do instead?*

An Intuition: Probability as Area

Draw a Venn diagram on the ground, with each area proportional to the number of elementary events it contains. Throw a stone into the rectangle at random.

The probability of A is how likely the stone lands inside A, assuming

1. we hit the rectangle Ω for certain, and
2. every part of Ω is equally likely to be hit.

Probability is then a **measure of the state** $A \subseteq \Omega$ of a random experiment — literally, how much of the sample space the event occupies.

Both assumptions are doing real work. Assumption 2 is the one we will have to give up.

Two Ways to Read a Probability

Frequentist
The probability of an elementary event a is the fraction of repetitions of the experiment that yield a.
Requires the experiment to be repeatable.

Bayesian
The probability of a is the degree of certainty that a will happen — even without conducting the experiment.
Requires a stated state of knowledge.

Both interpretations yield the same mathematics. They are mental models for what the number *means*, not different theories.

The Classical (Laplace) Definition

Definition (Classical / Laplace probability). If every elementary event in Ω is assigned the same probability, then

$$P(A) = \frac{|A|}{|\Omega|} = \frac{\text{number of cases favourable for } A}{\text{number of possible outcomes}}$$

An experiment whose configuration and procedure make outcomes equiprobable — by definition or by construction — is a **Laplace experiment**; its probabilities need not be estimated, they are implied by the setup.

Why That Definition Is Not Enough

> **The circularity.** "Laplace probability" defines probability using *equiprobability* — that is, using another notion of probability. Strictly speaking it is a circular definition.

The frequentist repair — defining probability as the limit of relative frequencies — fails too: such a limit exists only in mathematical settings, never in an experiment that can actually be repeated.

So what do we do? Stop trying to *define* what probability is, and instead specify what it must *do*.

The Kolmogorov Axioms

Definition (Probability measure, Kolmogorov 1933). Let Ω be a sample space and $\mathcal{P}(\Omega)$ its event space. A function $P : \mathcal{P}(\Omega) \to \mathbb{R}$ is a **probability measure** if:

Axiom I

$$P(A) \geq 0$$

non-negative

Axiom II

$$P(\Omega) = 1$$

normalised

Axiom III

$$A \cap B = \emptyset \Rightarrow$$
$$P(A \cup B) = P(A) + P(B)$$

additive

Nothing is said about *which* probabilities the events get — only about how any assignment must behave.

Beyond finite sample spaces. When Ω is infinite, the event space is not the full power set but a **σ-algebra** of measurable sets, and Axiom III is required for **countably many** disjoint events. Everything we compute in this course is unaffected; the construction belongs to measure theory.

Probability Space

> **Definition.** Let Ω be a sample space, $\mathcal{P}(\Omega)$ an event space, and P a probability measure. The triple $(\Omega, \mathcal{P}(\Omega), P)$ — also written (Ω, P) — is a **probability space**.

Every probabilistic statement in this course is made inside some probability space. Naming it is the discipline that keeps modelling assumptions visible.

What the Axioms Already Give Us

$$P(A) + P(\overline{A}) = 1$$

(Axioms II, III)

$$P(\emptyset) = 0$$

(from the above with $A = \Omega$)

$$A \subseteq B \Rightarrow P(A) \leq P(B)$$

monotonicity *(Axioms I, III)*

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

inclusion–exclusion *(Axiom III)*

For mutually exclusive $A_1, \ldots, A_p$: $P(A_1 \cup \ldots \cup A_p) = P(A_1) + \ldots + P(A_p)$.

Takeaway

> Probability is not defined by counting or by repetition — it is any assignment of numbers to events that is **non-negative, normalised, and additive**. The Laplace formula is one such assignment, not the definition.

Laplace Probabilities in Practice

When all outcomes in Ω are equally likely, computing $P(A)$ means **counting twice**: the favourable cases, and the possible ones.

$$P(A) = \frac{|A|}{|\Omega|}$$

The whole difficulty is choosing a sample space in which "equally likely" is actually true.

Example: Sum of Two Dice

What is the probability that two dice sum to 7?

Why ordered pairs? Because $(1, 6)$ and $(6, 1)$ are two distinct equally likely outcomes — the unordered sample space is **not** equiprobable.

Example: Sum of Two Dice

What is the probability that two dice sum to 7?

- $|\Omega| = 6 \cdot 6 = 36$ ordered pairs
- $E = \{(1,6), (6,1), (2,5), (5,2), (3,4), (4,3)\}$, so $|E| = 6$

$$P(\text{sum} = 7) = \frac{6}{36} = \frac{1}{6}$$

Why ordered pairs? Because $(1,6)$ and $(6,1)$ are two distinct equally likely outcomes — the unordered sample space is **not** equiprobable.

Example: Ordered or Unordered — Both Work

An urn holds **6 white** and **5 black** balls. Three are drawn. **What is the probability of exactly 1 white and 2 black?**

Variant 1 — order matters

$|\Omega| = 11 \cdot 10 \cdot 9 = 990$

Favourable: white-black-black, black-white-black, black-black-white $= 6 \cdot 5 \cdot 4 + 5 \cdot 6 \cdot 4 + 5 \cdot 4 \cdot 6 = 360$

$$\frac{360}{990} = \frac{4}{11}$$

Variant 2 — order ignored

$|\Omega| = \binom{11}{3} = 165$

Favourable: $\underbrace{\binom{6}{1}}_{\text{the white}} \cdot \underbrace{\binom{5}{2}}_{\text{the blacks}} = 6 \cdot 10 = 60$

$$\frac{60}{165} = \frac{4}{11}$$

The General Rule

> When selecting k items from n, you may count **with** or **without** order — as long as you are consistent, and as long as the chosen sample space really is equiprobable.

Pick whichever counting is easier for the event at hand.

Mini-Exercise

The urn again, smaller: **2 white and 2 black** balls, two drawn. What is the probability of **one of each**? Count without order, then check with order.

Mini-Exercise

The urn again, smaller: **2 white and 2 black** balls, two drawn. What is the probability of **one of each**? Count without order, then check with order.

Unordered: $\binom{4}{2} = 6$ pairs, $2 \cdot 2 = 4$ mixed, so $4/6 = 2/3$. Ordered: $4 \cdot 3 = 12$ draws, $2 \cdot 2 + 2 \cdot 2 = 8$ mixed, so $8/12 = 2/3$ — the same, because both sample spaces are equiprobable.

Takeaway

Laplace's formula — favourable over possible cases — is **one** probability measure, valid only in a sample space whose outcomes are equally likely. The formula is trivial; choosing that sample space is the work.

Module 4

Conditioning, Independence, Bayes

> ***Guiding Question:*** *A 99%-accurate test says you are sick. Why is that not a 99% chance of being sick — and what is the correct bookkeeping?*

Motivating Example

You roll a die twice. **After the first roll you see a 3.** Knowing this, what is the probability that both rolls sum to 8?

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You roll a die twice. **After the first roll you see a 3.** Knowing this, what is the probability that both rolls sum to 8?

- Before the first roll: $\Omega = \{(1,1),(1,2),\ldots,(6,6)\}$, $|\Omega| = 36$
- After seeing the 3, only $\Omega_{red} = \{(3,1),\ldots,(3,6)\}$ remains
- Outcomes like $(2,4)$ have become **impossible**
- The reduced space now carries all the probability: $P(\Omega_{red}) = 1$
- "Sum is 8" inside it: $E = \{(3,5)\}$

$$P = \frac{1}{6}$$

Definition

Definition (Conditional probability).

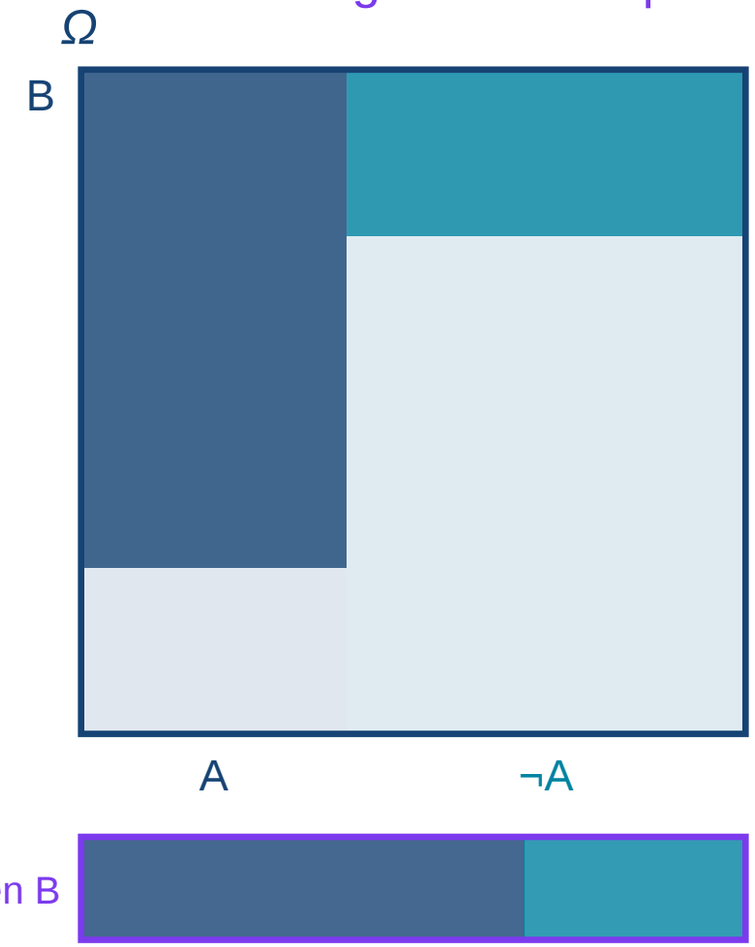
$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}, \qquad \text{if } P(B) > 0$$

$P(A \mid B)$ is the probability of A **under the condition** B.

Back to the stone: *given that the stone landed in area B, how likely is it that it also landed in A?* We renormalise by the area of B, because B is now the whole world.

Conditioning, Drawn

Conditioning on B = keep the B cells, stretch them back to width 1



P(A) = 0.40

P(B | A) = 0.75, P(B | ¬A) = 0.25

P(B) = 0.45

P(A ∩ B) = 0.30

P(A | B) = 0.67

Before the last click: what must the two shaded cells be divided by?

Example: Two Coins

Two coins are flipped. The event of interest is $B = \{(h, h)\}$.

(a) "The first coin is heads"

$F = \{(h, h), (h, t)\}$

$$P(B \mid F) = \frac{P(B \cap F)}{P(F)} = \frac{1/4}{1/2} = \frac{1}{2}$$

(b) "At least one coin is heads"

$A = \{(h, h), (h, t), (t, h)\}$

$$P(B \mid A) = \frac{P(B \cap A)}{P(A)} = \frac{1/4}{3/4} = \frac{1}{3}$$

Same event B, two different pieces of information, two different answers. **Why is (b) smaller?**

Total Probability

Theorem (Total probability). Let $B_1, \ldots, B_k$ be mutually exclusive with $\Omega = B_1 \cup \ldots \cup B_k$ and $P(B_i) > 0$. Then for any event A:

$$P(A) = \sum_{i=1}^{k} P(B_i) \cdot P(A \mid B_i)$$

Proof. Decompose A along the partition and use additivity:

$$P(A) = P((B_1 \cup \ldots \cup B_k) \cap A) = \sum_{i=1}^{k} P(B_i \cap A) = \sum_{i=1}^{k} P(B_i) \cdot P(A \mid B_i)$$

Rules That Follow

Product rule

$$P(A \cap B) = P(B) \cdot P(A \mid B)$$
$$= P(A) \cdot P(B \mid A)$$

Reversing the condition

$$P(B \mid A) = \frac{P(B) \cdot P(A \mid B)}{P(A)}$$

Complement, same condition

$$P(\overline{A} \mid B) = 1 - P(A \mid B)$$

But in general

$$P(A \mid \overline{B}) \neq 1 - P(A \mid B)$$

Held fixed at B, the map $A \mapsto P(A \mid B)$ satisfies the Kolmogorov axioms — conditioning yields a **new probability measure** P_B.

Chain Rule

$$P(E_1 \cap E_2 \cap \ldots \cap E_n) = P(E_1) \cdot P(E_2 \mid E_1) \cdot P(E_3 \mid E_1 \cap E_2) \cdots P(E_n \mid E_1 \cap \ldots \cap E_{n-1})$$

By successive application of the product rule.

> Any joint probability can be written as a chain of conditionals. Choosing a *good* order — and dropping conditions that do not matter — is exactly what graphical models will do later.

Takeaway

> Conditioning **shrinks the sample space and renormalises**. Everything else — product rule, total probability, Bayes, the chain rule — is bookkeeping on that single move.

Independence

Definition (Statistical independence). Two events A and B are **independent** iff

$$P(A \cap B) = P(A) \cdot P(B)$$

If additionally $0 < P(B) < 1$, this is equivalent to

$$P(A \mid B) = P(A) \qquad \text{and} \qquad P(A \mid B) = P(A \mid \overline{B})$$

Independence means: **learning B changes nothing about A.**

The Classic Confusion

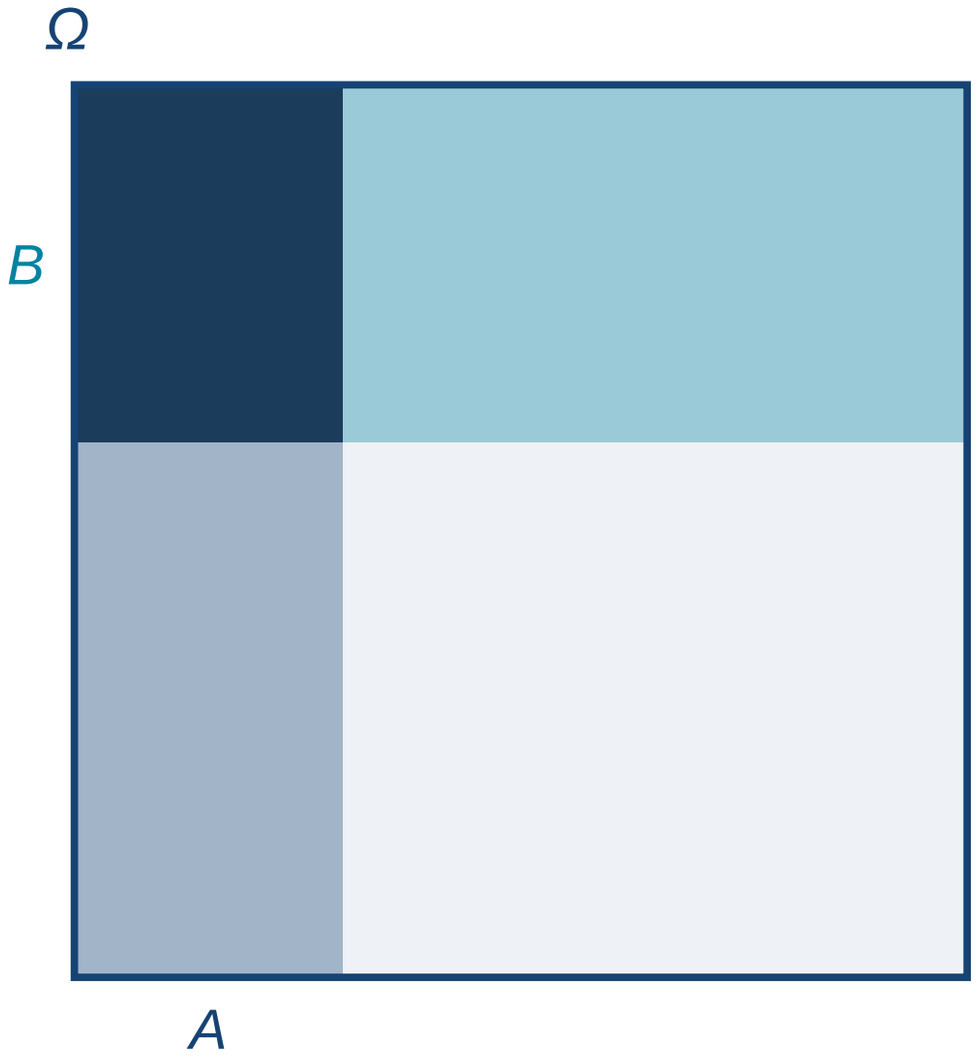
> **Statistical independence does not mean the events are disjoint.**

Disjoint events with positive probability are about as **dependent** as events can be: if A occurs, B certainly did not.

$$A \cap B = \emptyset \Rightarrow P(A \mid B) = 0 \neq P(A)$$

Independence Is About Areas, Not Overlap

$0.30 \cdot 0.40 = 0.12 = P(A \cap B) \rightarrow$ independent



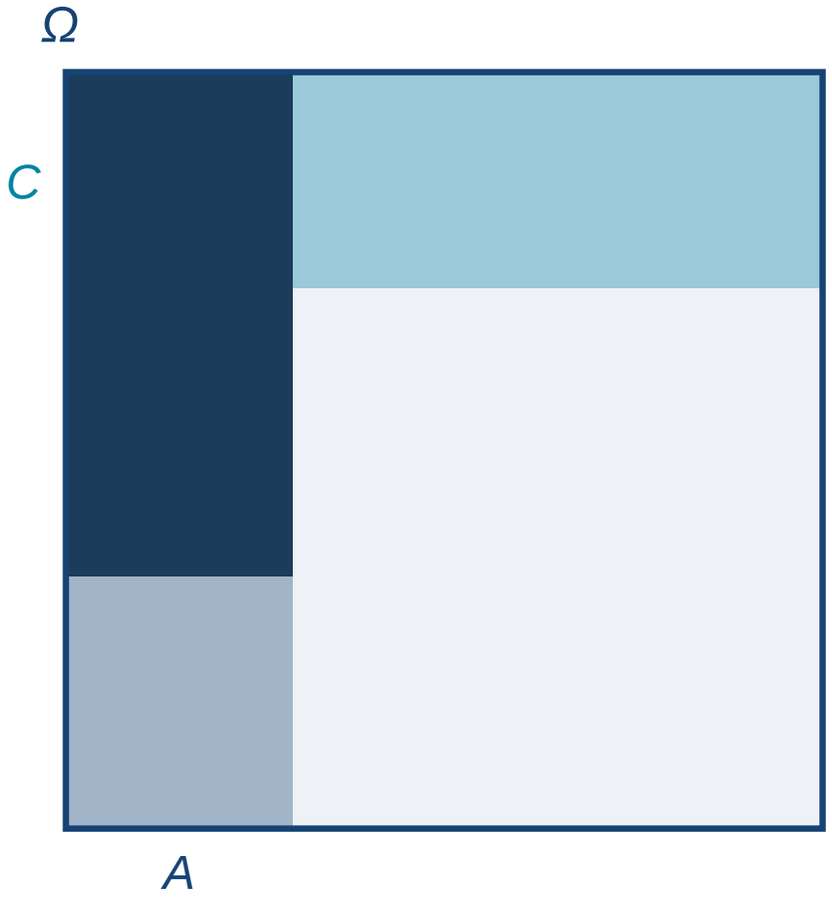
P(A) = 0.30

P(B) = 0.40

P(A ∩ B) = 0.12

The Same Picture, Dependent

$0.30 \cdot 0.40 = 0.12 \neq 0.20 = P(A \cap C)$ → dependent



P(A) = 0.30

P(C) = 0.40

P(A ∩ C) = 0.20

Here $P(A) \cdot P(C) = 0.12$ but $P(A \cap C) = 0.20$: the intersection is **larger** than independence would predict — A and C occur together more often than chance.

More Than Two Events

Events $A_1, \ldots, A_n$ are (mutually) independent iff for **every** subset of indices $i_1, \ldots, i_l$:

$$P(A_{i_1} \cap \ldots \cap A_{i_l}) = P(A_{i_1}) \cdots P(A_{i_l})$$

Pairwise independence is **not** enough — every subset must factorise.

Takeaway

> Independence is a statement about **numbers**, not pictures: $P(A \cap B)$ equals the product of the marginals. Disjoint is not independent, and identical marginals can hide either.

Conditional Independence

Definition. Events A and B are **conditionally independent given** C (written $A \perp\!\!\!\perp B \mid C$) iff

$$P(A \cap B \mid C) = P(A \mid C) \cdot P(B \mid C)$$

equivalently

$$P(A \mid B \cap C) = P(A \mid C)$$

Reading: **once C is known, B carries no further information about A.**

Dependent, but Independent Given C

Let $P(A) = \frac{8}{16}$, $P(B) = \frac{3}{16}$, $P(C) = \frac{4}{16}$ on a 16-cell sample space.

Unconditionally dependent

$$P(A \cap B) = \frac{1}{16}$$

$$P(A) \cdot P(B) = \frac{3}{32}$$

$\frac{1}{16} \neq \frac{3}{32}$, so $A \not\perp\!\!\!\perp B$

Conditionally independent given C

$$P(A \cap B \mid C) = \frac{1}{4}$$

$$P(A \mid C) \cdot P(B \mid C) = \frac{2}{4} \cdot \frac{2}{4} = \frac{1}{4}$$

so $A \perp\!\!\!\perp B \mid C$

Also directly: $P(A \mid B \cap C) = \frac{1}{2} = P(A \mid C)$.

Why This Matters

Common cause
A common cause C can make two effects look related; given C, they are independent.

Common effect
Two independent causes become dependent once their common effect is observed — *explaining away*.

Model size
Conditional independence lets a joint distribution factorise into small pieces — naive Bayes and graphical models are built on it.

Conditional independence neither implies nor is implied by ordinary independence. Both directions can fail.

Takeaway

> Conditioning can **create** independence as easily as it destroys it. "Given what I already know, does this extra fact still tell me anything?" is the question that makes large probabilistic models tractable.

Bayes' Theorem — An Intuition First

A test for a rare disease is **99% accurate**: positive for 99% of the sick, negative for 99% of the healthy. About **1 in 1,000 people** has the disease.

Your test comes back positive. How likely is it that you are actually sick?

Bayes' Theorem — An Intuition First

A test for a rare disease is **99% accurate**: positive for 99% of the sick, negative for 99% of the healthy. About **1 in 1,000 people** has the disease.

Your test comes back positive. How likely is it that you are actually sick?

> Only about **9%** — the answer most people miss by a factor of ten.

Why? Count People, Not Percentages

Restricting to the positives is exactly the conditioning step

P(sick | positive) = 1/11 ≈ 9%



What Just Happened

Restricting attention to the people who tested positive is exactly the move from the conditional-probability definition:

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}$$

The 11 positives are the new sample space B; the single sick person among them is $A \cap B$. The rarity of the disease survives the evidence because there were so many more healthy people to misclassify.

The trap

"99% accurate" describes $P(\text{positive} \mid \text{sick})$.

You wanted $P(\text{sick} \mid \text{positive})$.

These are different numbers.

Bayes' Theorem — Two Events

Theorem (Bayes, 2 events). Let $(\Omega, \mathcal{P}(\Omega), P)$ be a probability space and $A, B \in \mathcal{P}(\Omega)$ two events with $P(B) > 0$. Then:

$$P(A \mid B) = \frac{P(B \mid A) \cdot P(A)}{P(B)}$$

posterior — what we believe after

likelihood — how well A explains B

evidence — how ordinary B is

prior — belief before the evidence

Which of the four was "99% accurate", and which one did the intuition ignore?

Proof — Two Events

From the definition of conditional probability, twice:

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)} = \frac{\frac{P(A \cap B)}{P(A)} \cdot P(A)}{P(B)} = \frac{P(B \mid A) \cdot P(A)}{P(B)}$$

Bayes' Theorem — Multiple Hypotheses

Theorem (Bayes, multiple events). Let $A_i \in \mathcal{P}(\Omega)$ be mutually exclusive with $\Omega = A_1 \cup \ldots \cup A_k$ and $P(A_i) > 0$, and let B be an event with $P(B) > 0$. Then

$$P(A_i \mid B) = \frac{P(B \mid A_i) \cdot P(A_i)}{P(B)} = \frac{P(B \mid A_i) \cdot P(A_i)}{\sum_{j=1}^{k} P(B \mid A_j) \cdot P(A_j)}$$

The denominator is the **total probability** of the evidence: the same observation, explained by every competing hypothesis, weighted by how much we believed each one beforehand.

Proof — Multiple Hypotheses

As before, with the theorem of total probability supplying the denominator:

$$P(B) = \sum_{j=1}^{k} P(B \mid A_j) \cdot P(A_j)$$

Mini-Exercise

Same test, but now screening a **high-risk group** where 1 in 20 people is sick — the test is unchanged.

Estimate $P(\text{sick} \mid \text{positive})$ before computing it.

Mini-Exercise — Solution

Mini-Exercise — Solution

Out of 1,000: 50 sick → 49.5 true positives; 950 healthy → 9.5 false alarms.

$$P(\text{sick} \mid \text{positive}) = \frac{49.5}{49.5 + 9.5} = \frac{49.5}{59} \approx 84\%$$

The test did not change. The prior did. This is why screening decisions depend on who is screened.

Takeaway

Bayes' theorem is the arithmetic of changing your mind: **posterior ∝ likelihood × prior**. Evidence never speaks alone — it is always weighed against how plausible the hypothesis was to begin with.

Module 5

Random Variables and Distributions

> ***Guiding Question:*** *We rarely care which outcome occurred, only what it was worth. How do we move from outcomes to quantities?*

Motivation: We Rarely Care About the Outcome Itself

Toss three fair coins: $\Omega = \{H, T\} \times \{H, T\} \times \{H, T\}$, eight outcomes.

Usually the interesting quantity is not *which* sequence occurred, but the **number of heads** Y:

$P(Y = 0) = \frac{1}{8}$	$P(Y = 1) = \frac{3}{8}$	$P(Y = 2) = \frac{3}{8}$	$P(Y = 3) = \frac{1}{8}$
(T, T, T)	$(H, T, T), (T, H, T), (T, T, H)$	$(H, H, T), (T, H, H), (H, T, H)$	(H, H, H)

Y compresses eight outcomes into four numbers — and the probabilities come along automatically.

Definition

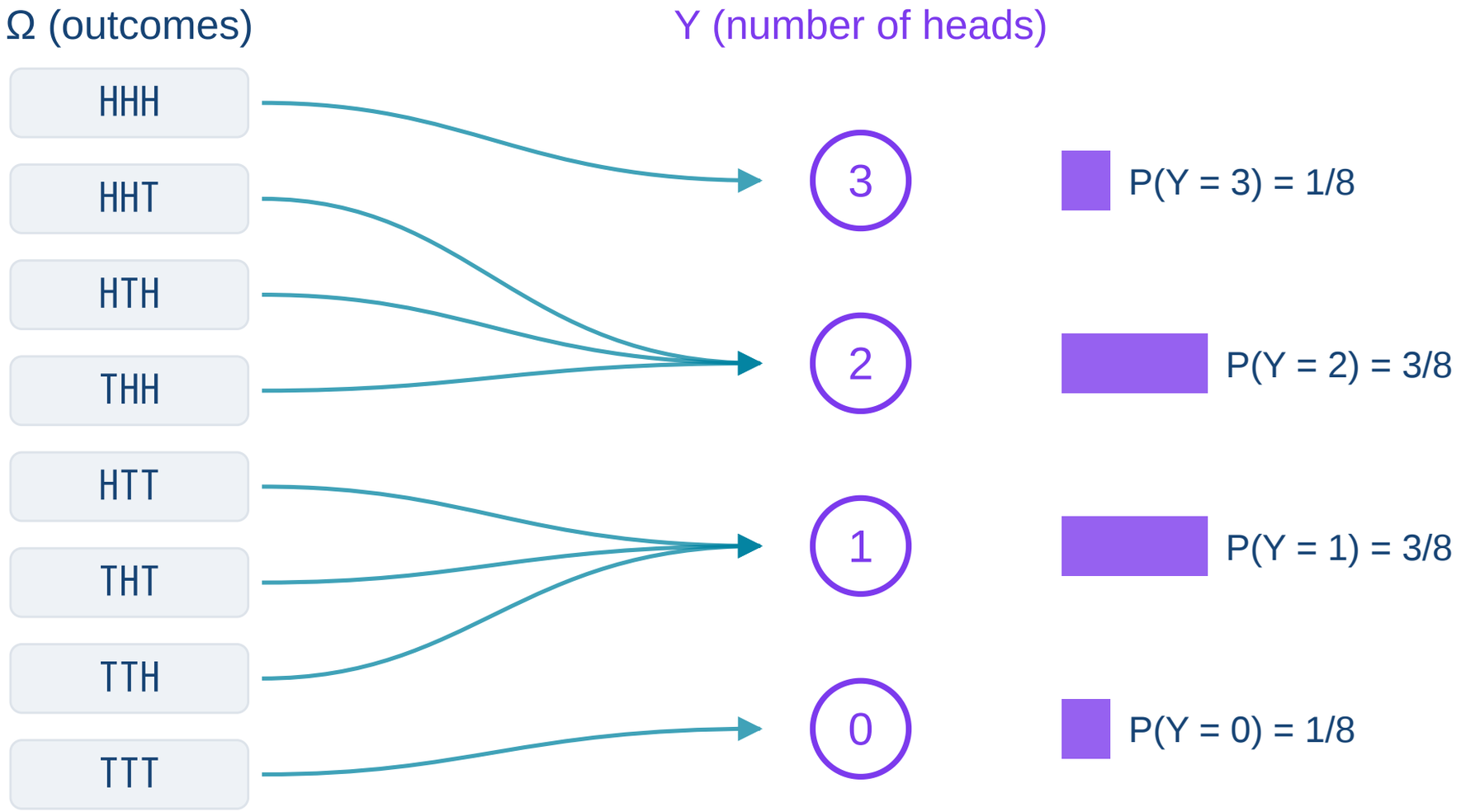
Definition (Random variable). A random variable X maps outcomes to a measurable range E: $X : \Omega \to E$. Its probabilities are inherited from the underlying probability space:

$$P(X \in S) = P(\{\omega \in \Omega \mid X(\omega) \in S\})$$

A random variable is neither random nor a variable — it is a **function** on the sample space. The randomness lives in ω, not in X.

The Function, Drawn

Three arrows arrive at 1 and at 2 — which is the whole shape



Examples

Two coins

$\Omega = \{HH, HT, TH, TT\}$

X = number of heads

People

$\Omega = \{\text{person}_1, \text{person}_2, \ldots\}$

X = height of the person

Two dice

$\Omega = \{\langle n_1, n_2 \rangle\}$

$X(\omega) = n_1 + n_2$

Discrete and Continuous

Discrete random variables

Countable range, usually $E \subseteq \mathbb{N}$.

- number of children of a person
- sum of two dice

Continuous random variables

Real-valued range, $E = \mathbb{R}$.

- height of a person
- horsepower of a car

The distinction decides *how* we may describe the probabilities — with a mass function or with a density.

A Random Variable With Infinitely Many Values

Flip a biased coin ($P(H) = p$) until the first head. Let X be the number of flips needed:

$$P(X = 1) = p, \quad P(X = 2) = (1-p)p, \quad P(X = 3) = (1-p)^2 p$$

$$P(X = n) = (1-p)^{n-1} p$$

Countably infinite range, and the probabilities still sum to 1.

Takeaway

A random variable is a **function from outcomes to numbers** that carries the probability measure with it. It lets us stop talking about sample spaces and start talking about quantities — which is what the rest of the course does.

Probability Mass Function

Definition (PMF). For a discrete random variable X, the **probability mass function** is

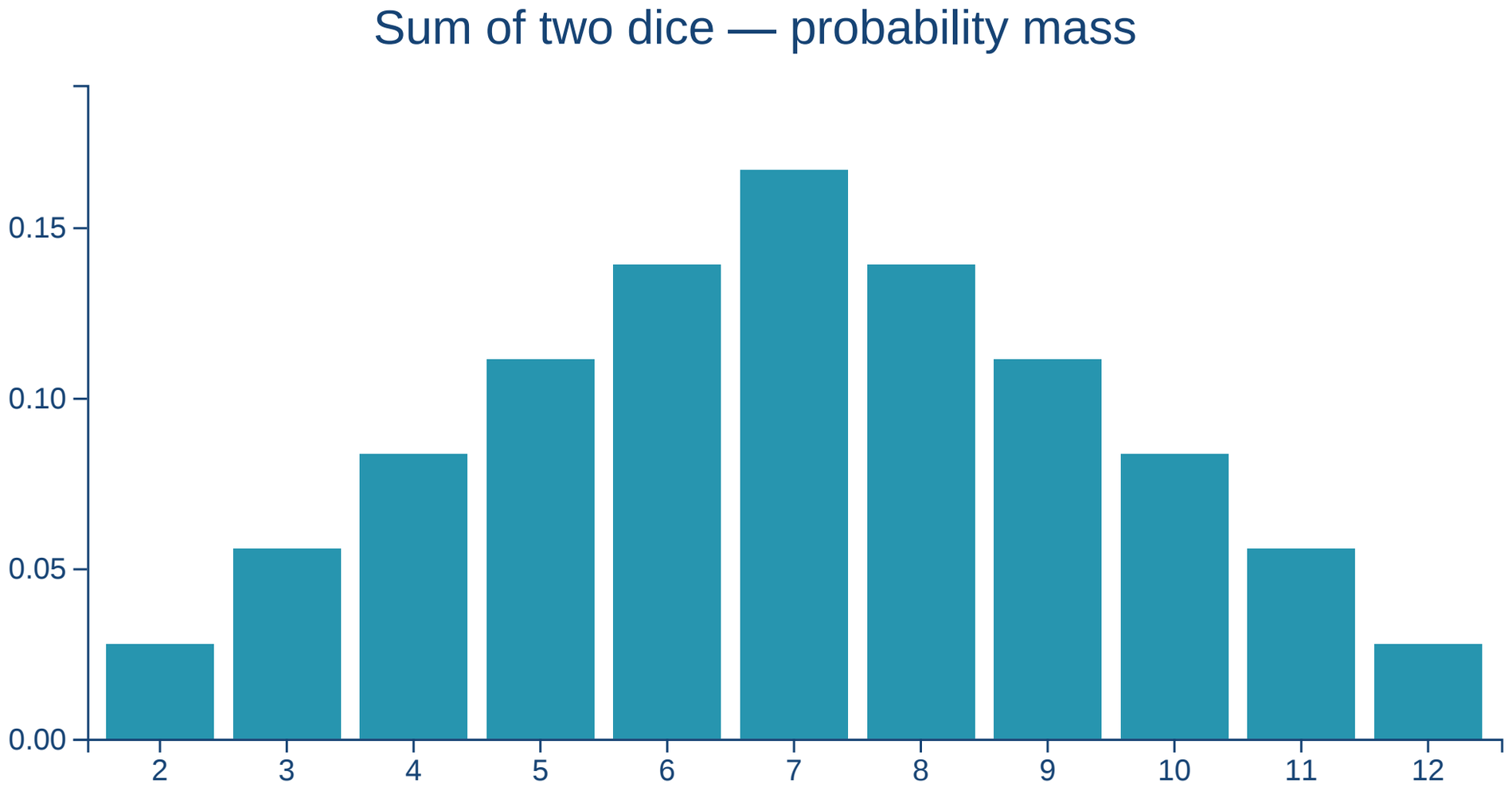
$$p(a) = P(X = a)$$

$p(a) > 0$ for at most countably many a

$\sum_i p(a_i) = 1$

p is itself a probability measure

Example: Sum of Two Dice



The triangular shape is counting, drawn: six of the 36 outcomes give 7, only one gives 2.

Expectation

Definition (Expected value).

$$E[X] = \sum_i a_i \cdot p(a_i)$$

- Frequentist reading: the **average outcome** if the experiment were repeated infinitely often
- The first moment: $\mu = E[X^1]$; the first *central* moment $E[X - E[X]]$ is always 0

For a real function g:

$$E[g(X)] = \sum_i g(a_i)p(a_i)$$

and for constants a, b:

$$E[aX + b] = aE[X] + b$$

Variance

Definition (Variance).

$$\mathrm{Var}(X) = E\big[(X-\mu)^2\big] = E[X^2] - \big(E[X]\big)^2$$

- a measure of **dispersion** around the mean — larger variance, larger typical deviation
- the second central moment
- $\mathrm{Var}(aX+b) = a^2\mathrm{Var}(X)$
- $\sigma = \sqrt{\mathrm{Var}(X)}$ is the **standard deviation**, in the same unit as X

Takeaway

A discrete distribution is a PMF; expectation and variance compress it into **where it sits** and **how much it spreads**.

Bernoulli and Binomial

Bernoulli(p) — one trial

$X = 1$ success, $X = 0$ failure

$$E[X] = p, \quad \mathrm{Var}(X) = p(1-p)$$

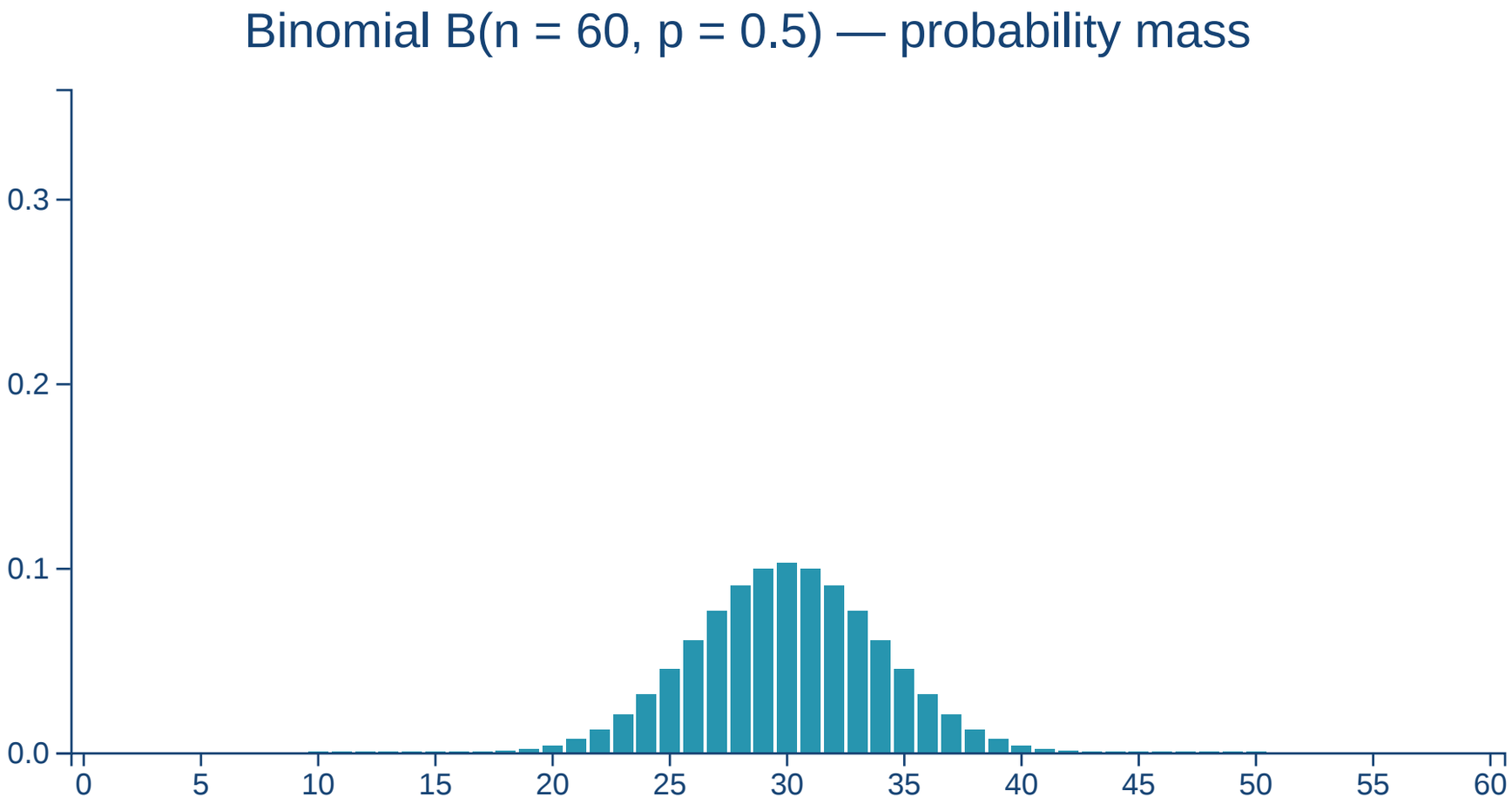
Binomial $B(n, p)$ — n independent trials

$$p(k) = \binom{n}{k} p^k (1-p)^{n-k}$$

$$E[X] = np, \quad \mathrm{Var}(X) = np(1-p)$$

$\binom{n}{k} = \frac{n!}{k!\,(n-k)!}$ counts the sequences with k successes; the PMF multiplies it by the probability of one such sequence — combinatorics and independence in a single line.

How the Shape Depends on the Parameters



Predict before each click: as n grows, does the distribution get wider or narrower — and what happens to its shape?

Mini-Exercise

A student guesses all **10** questions of a quiz, each with **4** options.

How many correct answers should they expect — and is 5 or more likely?

Mini-Exercise

A student guesses all **10** questions of a quiz, each with **4** options.

How many correct answers should they expect — and is 5 or more likely?

$B(10, 0.25)$: on average $10 \cdot 0.25 = 2.5$ correct. Five or more is twice the expectation — unlikely (about 8%).

Takeaway

A yes/no attribute is a **Bernoulli** variable with one parameter p; counting the yeses in n independent trials gives a **binomial**. Both are counting in probabilistic clothing — and p is the number data will have to tell us.

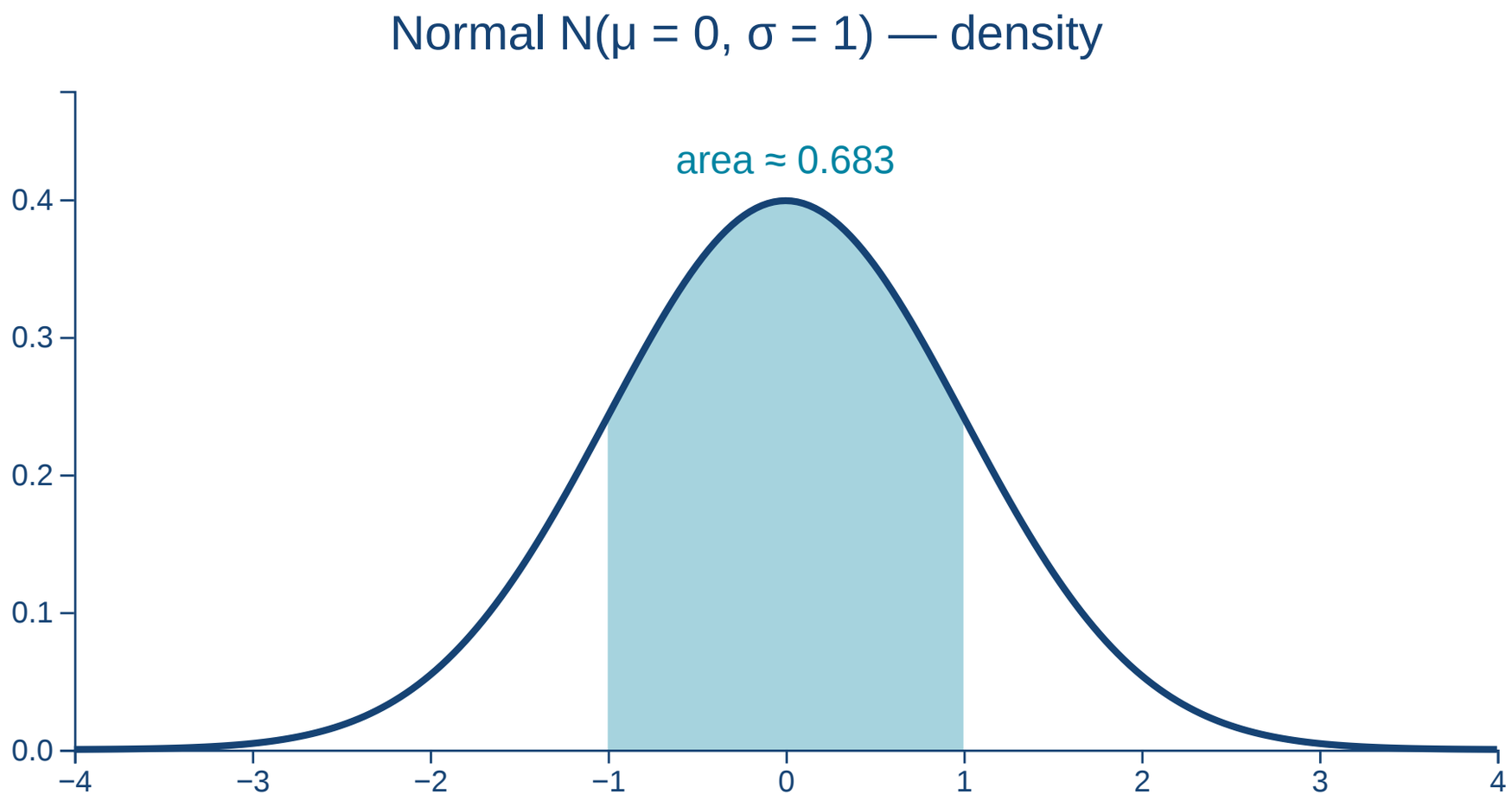
From Mass to Density

Definition (PDF). A continuous random variable X has a **probability density function** f with

$$P(b \leq X \leq c) = \int_b^c f(a)da, \qquad \int_{-\infty}^{\infty} f(a)da = 1$$

$f(a)$ is **not** a probability — it is probability *per unit interval*. Consequently $f(a) > 1$ is perfectly possible, and $P(X = a) = 0$ for every single value a.

Probability Is Area Under the Curve



The shaded region is $P(-1 \le X \le 1)$ — the same "probability as area" intuition from the axioms unit, now with a curve instead of a Venn diagram.

Expectation and Variance — Sums Become Integrals

Discrete

$$E[X] = \sum_i a_i p(a_i)$$

$$E[g(X)] = \sum_i g(a_i) p(a_i)$$

Continuous

$$E[X] = \int_{-\infty}^{\infty} a f(a) da$$

$$E[g(X)] = \int_{-\infty}^{\infty} g(a) f(a) da$$

The rules are unchanged: $E[aX + b] = aE[X] + b$, and $\mathrm{Var}(X) = E[(X - \mu)^2]$.

Two Basic Continuous Distributions

Uniform on $[a, b]$

$$f(x) = \frac{1}{b-a} \text{ for } x \in [a, b]$$

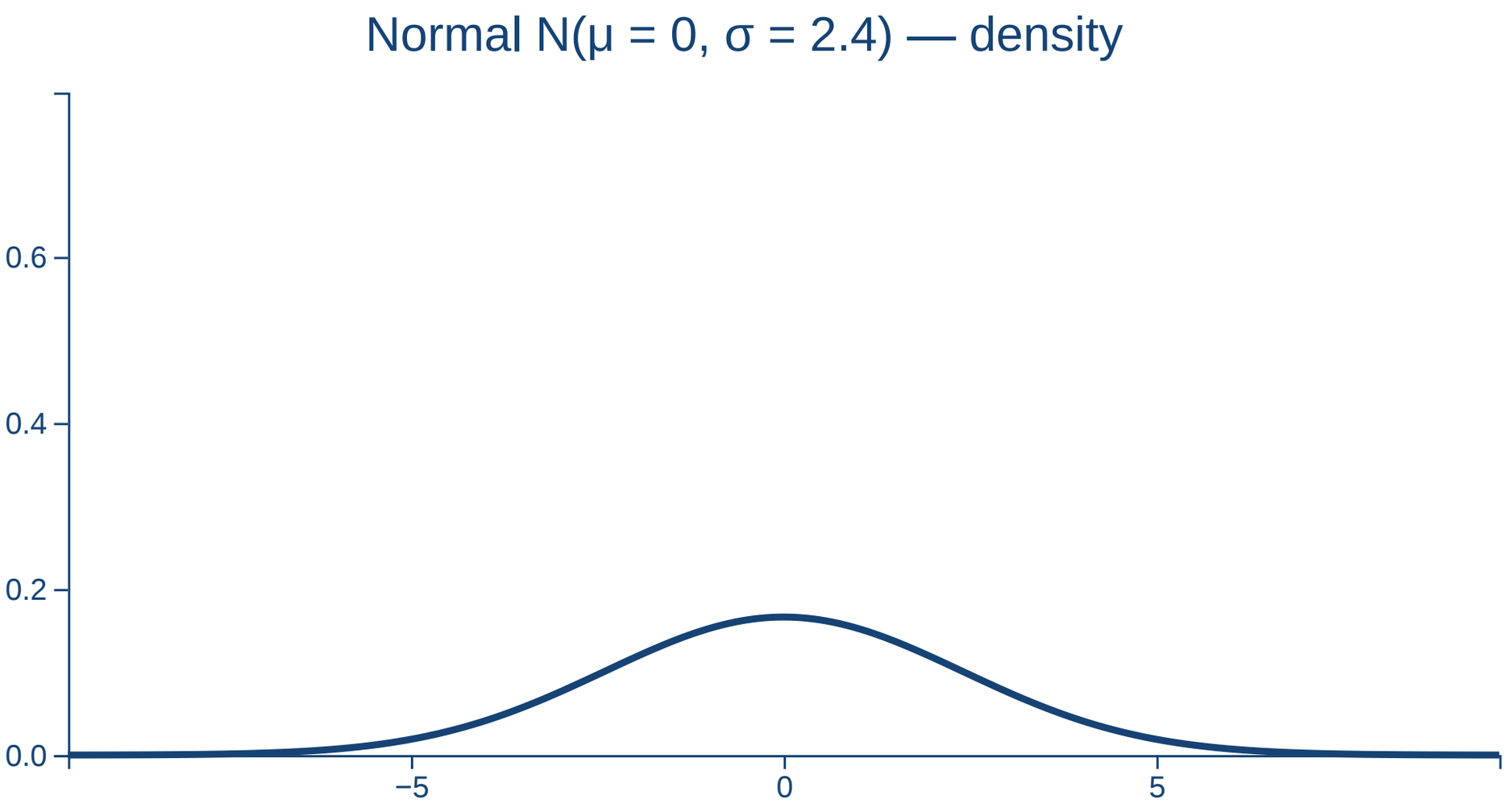
Every interval of equal length is equally likely — the Laplace assumption, continuous version.

Normal $N(\mu, \sigma^2)$

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

$E[X] = \mu$, $\mathrm{Var}(X) = \sigma^2$; $N(0, 1)$ is the **standard** normal.

What σ Does to the Shape



The peak falls as the curve widens — the area is always 1.

Why the Normal Is Everywhere

> The **central limit theorem**: under mild conditions, the average of n independent observations is approximately normal for large n — whatever the original distribution was.

This is why the normal is the default when the true distribution is unknown — and why a sum of many independent coin flips, the binomial, looks bell-shaped.

Other Distributions You Will Meet

- **Exponential** — time until an event occurs (e.g. until the next earthquake)
- **Gamma** — time until n events have occurred
- **Beta** — values in $[0, 1]$, e.g. an unknown probability
- **Dirichlet** — distributions *over* probabilities

Continuous distributions replace mass with **density** and sums with **integrals**. Probability remains area, expectation remains a weighted average, and the normal is the default shape because of the central limit theorem.

More Than One Random Variable

Univariate	Bivariate	Multivariate
one random variable	two random variables	three or more

A **joint distribution** describes the outcomes of several random variables *at the same time* — which is what every dataset with more than one column is.

Joint Mass and Density

Discrete — joint PMF

$$p_{X,Y}(a,b) = P(X = a, Y = b)$$

Continuous — joint PDF

$$P((X,Y) \in R) = \iint_R f_{X,Y}(a,b)dadb$$

Notational convention from here on. We stop distinguishing discrete from continuous variables and simply write $p(a,b,c,\ldots)$ for the probability that $\{X_1, X_2, X_3, \ldots\}$ is in state $a, b, c, \ldots$

Marginalisation

Marginalisation eliminates a random variable by summing over all its states. For $p(x_1, \ldots, x_n)$, a **marginal** over a subset is

$$p(x_1) = \sum_{x_2} \cdots \sum_{x_n} p(x_1, x_2, \ldots, x_n)$$

"Forget about a variable, but keep its influence correctly accounted for."

Example: Boys and Girls

15% of families have no children, 20% have one, 35% have two, 30% have three. Each child is independently a boy or a girl with probability $\frac{1}{2}$.

Let B = number of boys, G = number of girls.

$p(B, G)$	$G = 0$	$G = 1$	$G = 2$	$G = 3$	$p(B)$
$B = 0$	.15	.10	.0875	.0375	.375
$B = 1$	.10	.175	.1125	0	.3875
$B = 2$	.0875	.1125	0	0	.2
$B = 3$	.0375	0	0	0	.0375
$p(G)$	.375	.3875	.2	.0375	1

The margins of the table *are* the marginals — which is where the name comes from.

Covariance

$$\operatorname{cov}(A, B) = E\big[(A - E[A])(B - E[B])\big]$$

$\operatorname{cov} > 0$
A large when B large — positively related

$\operatorname{cov} < 0$
A large when B small — negatively related

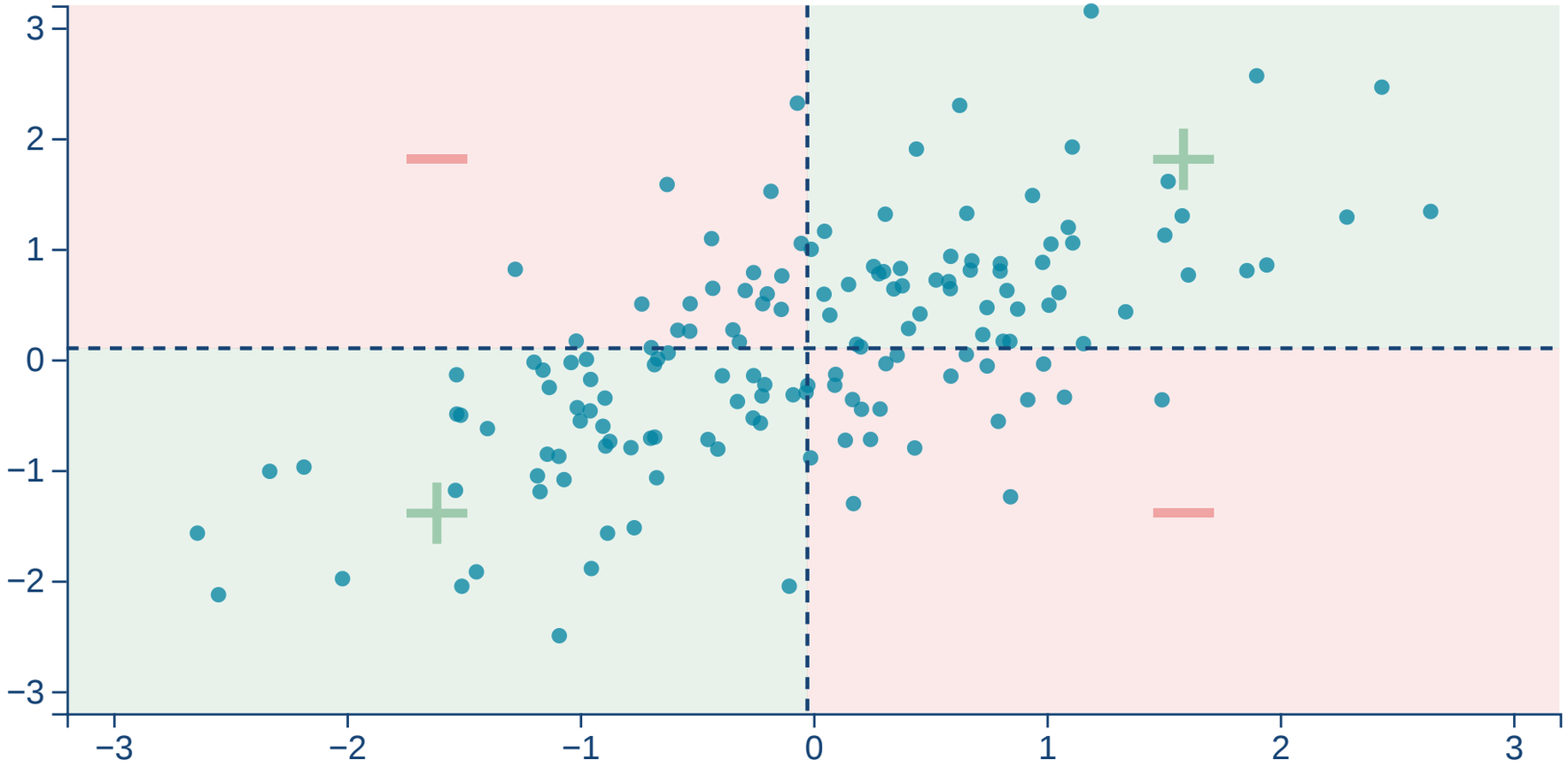
$\operatorname{cov}(A, A) = \operatorname{Var}(A)$
variance is self-covariance

Independence implies zero covariance. **The converse is false** — covariance only detects *monotone linear* association.

Covariance Sees Only One Kind of Structure

Slide ρ, then switch the relation. Predict the covariance first.

cov(X, Y) = 0.62



Predict first: with $Y = X^2 - 1$, every point is fixed by its x. The covariance?

Takeaway

The joint distribution holds everything; **marginalisation** sums a variable away, **conditioning** restricts, and **covariance** summarises linear co-variation. Zero covariance is weaker than independence.

Module 6

Describing Data

> ***Guiding Question:*** *The distribution is unknown and all we have is a sample. What can we honestly say about it?*

We Never Know the Distribution

Up to now we assumed a distribution was given. In practice it is **unknown**, and we must estimate its properties from data.

Population
The entire group — all humans on earth, all transactions, all documents.
Its properties (μ, σ^2) are the truth we cannot observe.

Sample
The observations we actually have.
Its properties ($\overline{x}$, s^2) are what we compute; estimates of the population's ($\overline{x}$, $\hat{\sigma}^2$) are built from them.

Descriptive statistics summarise a sample, usually while assuming something about the underlying distribution.

What We Summarise

Central tendency

1st order moment
sample mean, median

Dispersion

2nd central moment
variance, standard deviation, interquartile range

Shape

3rd central moment
skewness — or, without moments, quantiles

Univariate statistics describe one random variable; **multivariate** statistics describe $X_1, \ldots, X_d$ together.

Means

Sample mean

$$\overline{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$$

Weighted mean

$$\overline{x}_w = \frac{\sum_i w_i x_i}{\sum_i w_i}$$

Geometric mean — for rates and ratios

$$\left(\prod_i x_i\right)^{1/n}$$

Harmonic mean — for averaging rates

$$\frac{n}{\sum_i 1/x_i}$$

Trimmed mean: drop the k smallest and largest values, then average — a cheap way to buy robustness.

Median

The **median** is the middle value of the sorted sample: for odd n the $\frac{n+1}{2}$-th value, for even n the average of the two middle values.

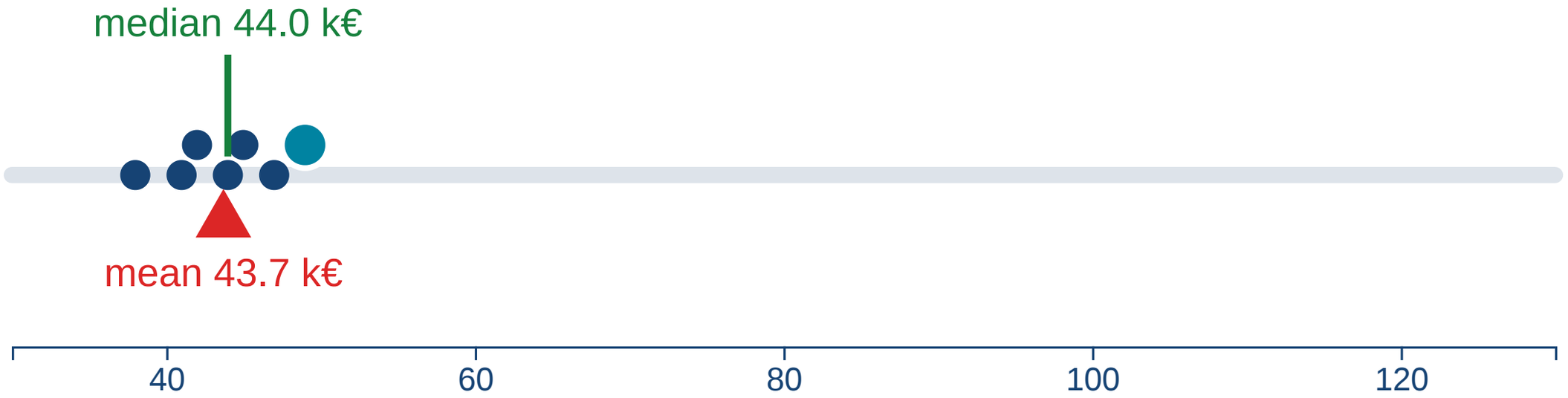
Robust against outliers — one extreme value moves the median by at most one position

Costlier to compute: requires sorting, or an interval approximation over binned data

For binned data, the median is interpolated within its interval from the cumulative counts, the interval's lower border and width.

One Value, Two Very Different Reactions

Drag the light point — which marker follows it?



Commit first: drag the top salary to the far right. Which marker follows?

Mini-Exercise

Salaries in a small firm (in k€): 38, 41, 42, 44, 45, 47, 260.

Compute the mean and the median. Which number should appear in a recruitment ad, and which one describes a typical employee?

Mini-Exercise — Solution

Mini-Exercise — Solution

Mean ≈ 73.9, median $= 44$.

The single large value pulls the mean above **every** salary except one. The median describes the typical employee; the mean describes the payroll divided by headcount. Both are correct — they answer different questions.

Dispersion

Dispersion describes how stretched or squeezed the data is around its centre.

Sample variance — the variance *of the observed data*

$$s^2 = \frac{1}{n}\sum_{i=1}^{n}(x_i - \overline{x})^2$$

Standard deviation $s = \sqrt{s^2}$, in the unit of the data.

Computable in **one pass** over the data, $O(n)$, using

$$\sum x_i^2 \text{ and } \sum x_i$$

— which matters when the data does not fit in memory.

Why the Denominator Changes

The **population variance** σ^2 is the variance over the whole population. If the true mean μ is unknown, we must estimate it — and using $\overline{x}$ in its place makes the deviations systematically too small.

$$\hat{\sigma}^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \overline{x})^2$$

Dividing by $n-1$ compensates: one degree of freedom was spent estimating the mean from the same data.

Naming is not uniform: many textbooks, and pandas, call the $n-1$ version s^2; NumPy's `var` divides by n unless told otherwise.

Takeaway

s^2 describes **the sample**; $\hat{\sigma}^2$ estimates **the population**. The $n-1$ is not a convention — it is the price of having estimated the mean from the same data.

Quantiles

> The k-th q-**quantile** is the value x such that k/q of the values are smaller and $(q-k)/q$ are larger ($0 < k < q$).

Percentiles

$q = 100$

"the 12th percentile"

Quartiles

$q = 4$

Q_1 = 25th, Q_2 = 50th (= median),
Q_3 = 75th percentile

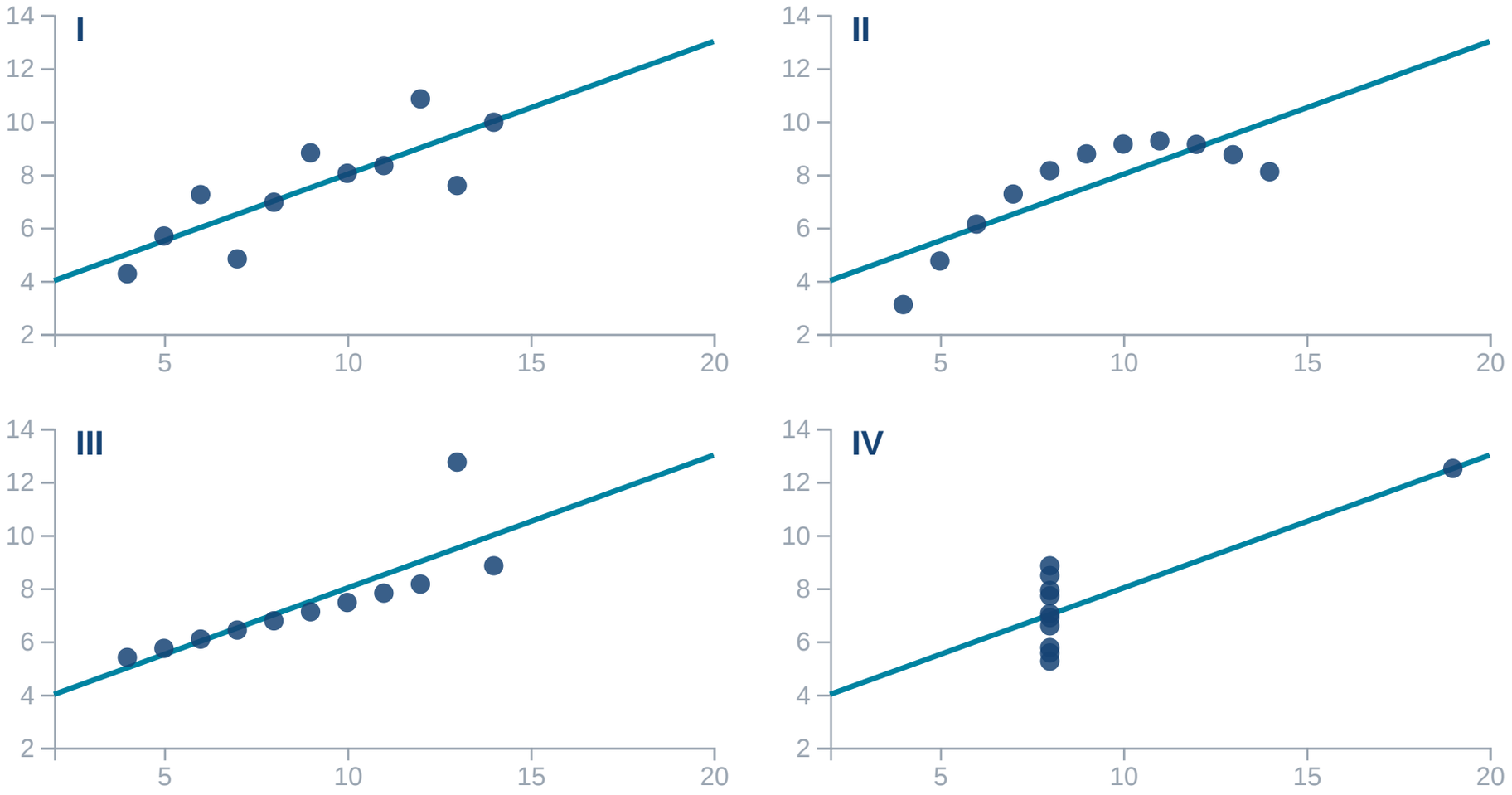
IQR

$$\text{IQR} = Q_3 - Q_1$$

the middle half of the data

Quantiles summarise **shape and skewness** without assuming a distribution.

Anscombe's Quartet



Same mean, same variance, same correlation, same regression line — different data

all four: $\bar{x} = 9$, $s^2_x = 11$, $\bar{y} = 7.50$, $s^2_y \approx 4.12$, $r \approx 0.82$, $\hat{y} = 3.00 + 0.500 \cdot x$

What the Quartet Teaches

Summaries compress — and compression loses.
Identical moments can hide a curve, an outlier, or a variable that barely varies at all.

Plot first, summarise second.
Visual inspection is not a soft alternative to statistics; it is how you find out which statistics are meaningful.

Quantiles describe a distribution without assuming its shape — and no set of summary numbers can replace looking at the data.

You Computed a Number. Which Number Did You Want?

Seven salaries gave a sample mean of $\overline{x} = 73.9$ k€.

What is the probability that the population mean μ is exactly 73.9 k€?

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Zero. $\overline{x}$ is not μ — it is one **estimate** of it, computed from the seven people who happened to be in the sample.

The honest question is not *what is* μ but **how far from** μ **can this number be?**

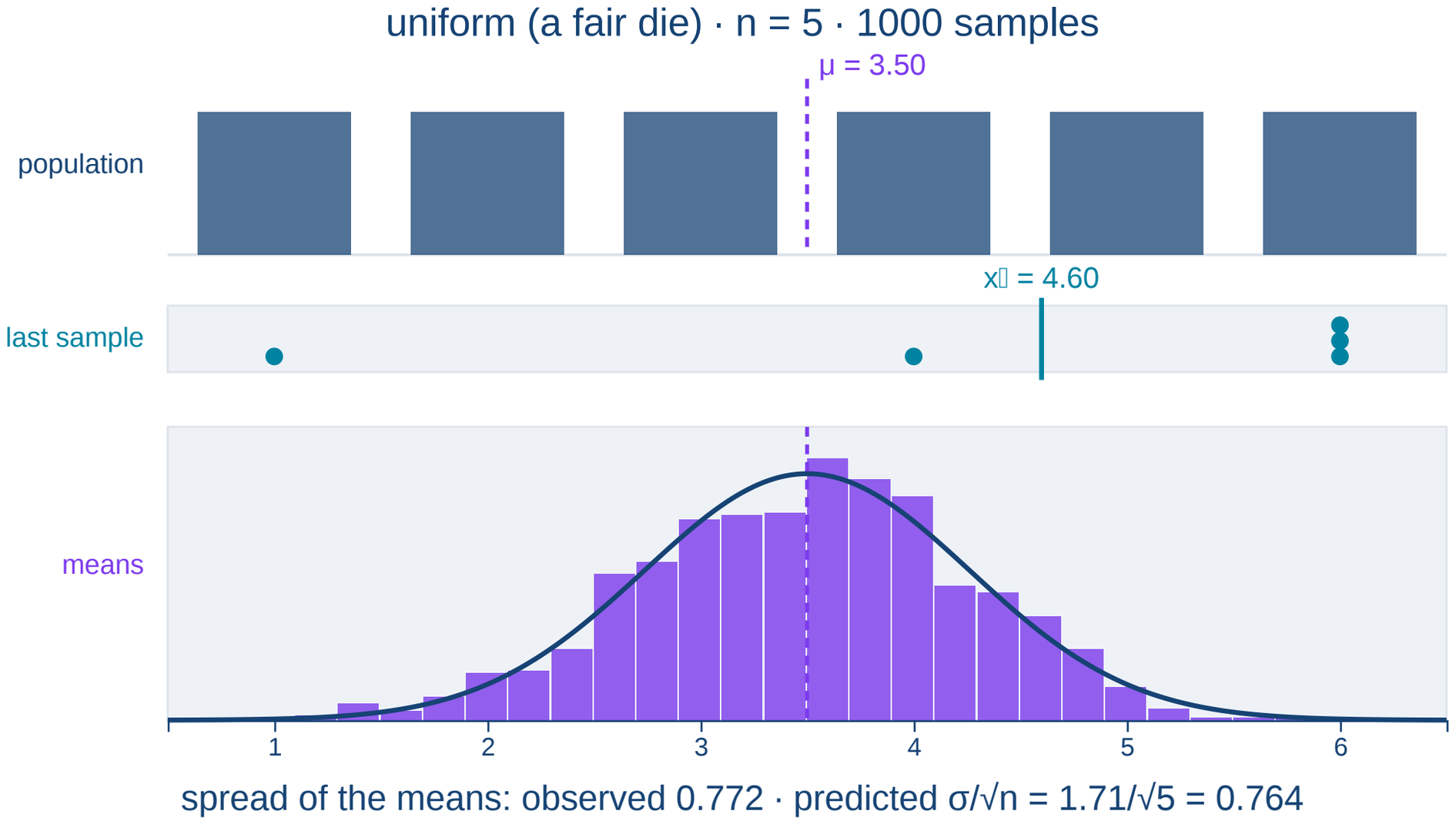
Estimator and Estimate

Definition. An **estimator** is a function of the sample — $\hat{\theta} = g(X_1, \ldots, X_n)$ — used to approximate an unknown quantity θ of the population. The number it takes on for one particular sample is an **estimate**.

The sample is random, so the estimator is a **function of random variables**: $\hat{\theta}$ is itself a random variable, with a distribution of its own.

$\overline{x}$, s^2 and $\hat{\sigma}^2$ from descriptive statistics are all estimators. We used them without saying what kind of object they are.

An Estimator Has a Distribution



Three Things the Picture Shows

- → The means pile up **around** μ — the estimator is not systematically off
- → The pile is **narrower than the population** — averaging cancels noise
- → The pile is **bell-shaped**, even though the population was not

> **Definition (Sampling distribution).** The distribution of an estimator $\hat{\theta}$ over all possible samples of size n. $\hat{\theta}$ is **unbiased** if $E[\hat{\theta}] = \theta$.

Use the Widget: Change the Population

Set the population to **skewed** and then to **bimodal**, keeping $n = 5$.

Which of the three observations survive a population that is not symmetric?

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All three. The shape of the sampling distribution stops depending on the shape of the population as soon as n is more than a handful — which is why one formula can cover cases we never inspect.

Standard Error

Definition (Standard error). The standard deviation of an estimator's sampling distribution. For the sample mean of independent draws:

$$\mathrm{SE}(\overline{X}) = \frac{\sigma}{\sqrt{n}}$$

It is measured in the unit of the data and answers "how far off is a typical estimate", where σ answers "how far apart are typical *observations*".

σ and SE are different quantities. More data does nothing to σ — the world stays as variable as it was — and shrinks SE.

Precision Costs Data

Multiply the data by…	…and the standard error is multiplied by
2	0.71
4	**½**
25	⅕
100	**1⁄10**

The $\sqrt{n}$ is why the last digit of an estimate is so expensive: halving the error costs four times the data, and a tenfold improvement costs a hundredfold.

Two Theorems We Have Already Seen Happen

Law of large numbers

$\overline{X}_n \to \mu$ as $n \to \infty$.

The estimate converges on the truth — the pile keeps narrowing.

Central limit theorem

$\overline{X}_n$ is approximately $N(\mu, \sigma^2/n)$ for large n.

And we know the *shape* of what is left.

The first says the error goes away; the second says how much of it is left after n observations. Only the second lets us put a number on our ignorance.

From a Number to an Interval

If $\overline{X} \approx N(\mu, \sigma^2/n)$, then for 95% of samples $\overline{X}$ lands within $1.96\,\mathrm{SE}$ of μ — and therefore μ lands within $1.96\,\mathrm{SE}$ of $\overline{X}$.

$$\overline{x} \pm 1.96 \cdot \frac{\sigma}{\sqrt{n}}$$

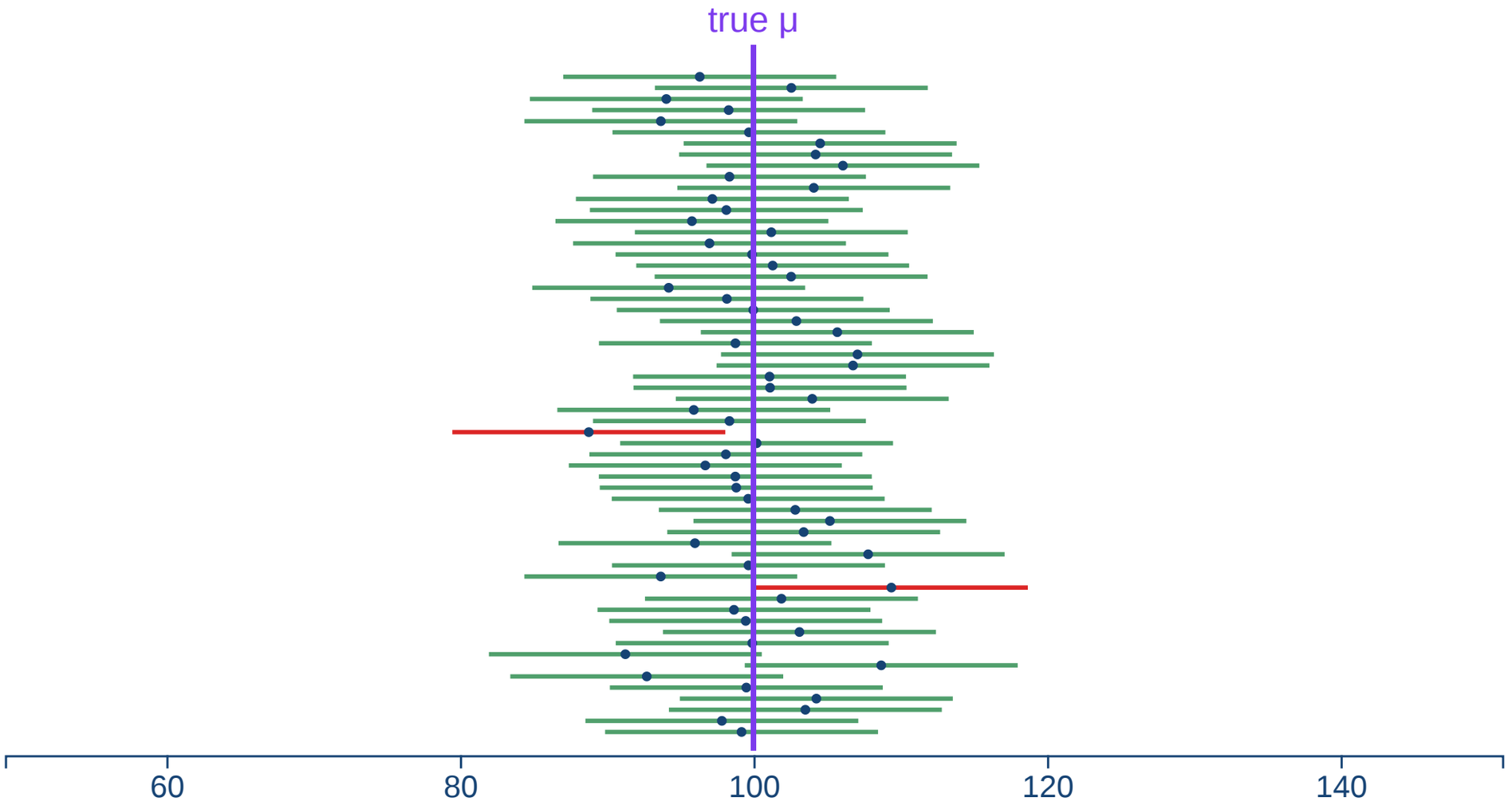
the estimate — our best single guess

how confident we insist on being

how noisy the estimate is

What Does "95% Confident" Mean?

Change the level or n — which number follows, and which does not?



58 of 60 intervals cover μ (97%) — nominal 95%

The Sentence Almost Everyone Gets Wrong

WHAT IT DOES NOT MEAN

"There is a 95% probability that μ lies in **this** interval."

μ is a fixed number. It is either in this interval or it is not — there is no probability left to have.

WHAT IT MEANS

"95% of the intervals produced **by this procedure** contain μ."

The probability is a property of the recipe, evaluated before the data is seen.

Where This Leaves Us

Two kinds of uncertainty meet in one formula. We can now measure one of them.

- → σ — how variable the world is. **Aleatoric.** Unchanged by data.
- → $\mathrm{SE} = \sigma/\sqrt{n}$ — how badly we know our own estimate. **Epistemic.** Shrinks with data.

"More data helps" was a slogan.
It is now a rate: $1/\sqrt{n}$.

Mini-Exercise

A shop timed $n = 100$ customers at a counter: $\overline{x} = 180$ s, and past records give $\sigma = 60$ s. What is the standard error — and how many customers would halve it?

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A shop timed $n = 100$ customers at a counter: $\overline{x} = 180$ s, and past records give $\sigma = 60$ s. What is the standard error — and how many customers would halve it?

$\mathrm{SE} = 60/\sqrt{100} = 6$ s, so the 95% interval is about 180 ± 12 s. Halving the SE needs twice the $\sqrt{n}$ — **four times the data**, $n = 400$. $\sigma = 60$ s does not change: customers stay as variable as they were; only our knowledge of their average improves.

Takeaway

A statistic is a **random variable**. Its distribution over repeated samples is the sampling distribution; its spread is the **standard error**, $\sigma/\sqrt{n}$. A confidence interval reports that spread — and its percentage describes the **procedure**, not the one interval in front of you.

Module 7

Hypothesis Testing

> ***Guiding Question:*** *Your new model scores higher than the baseline. How do you know the improvement is real?*

Motivation: Comparing Two ML Experiments

Experiment A — baseline model accuracy measured over n_A runs

Experiment B — new architecture accuracy measured over n_B runs

B scored higher. Is the improvement systematic, or just randomness?

H_0: the two means are equal — $\mu_A = \mu_B$
H_1: they differ — $\mu_A \neq \mu_B$

The Idea

> Compare what we **observed** with what H_0 **predicts**. If the disagreement is large enough to be implausible under H_0, reject H_0.

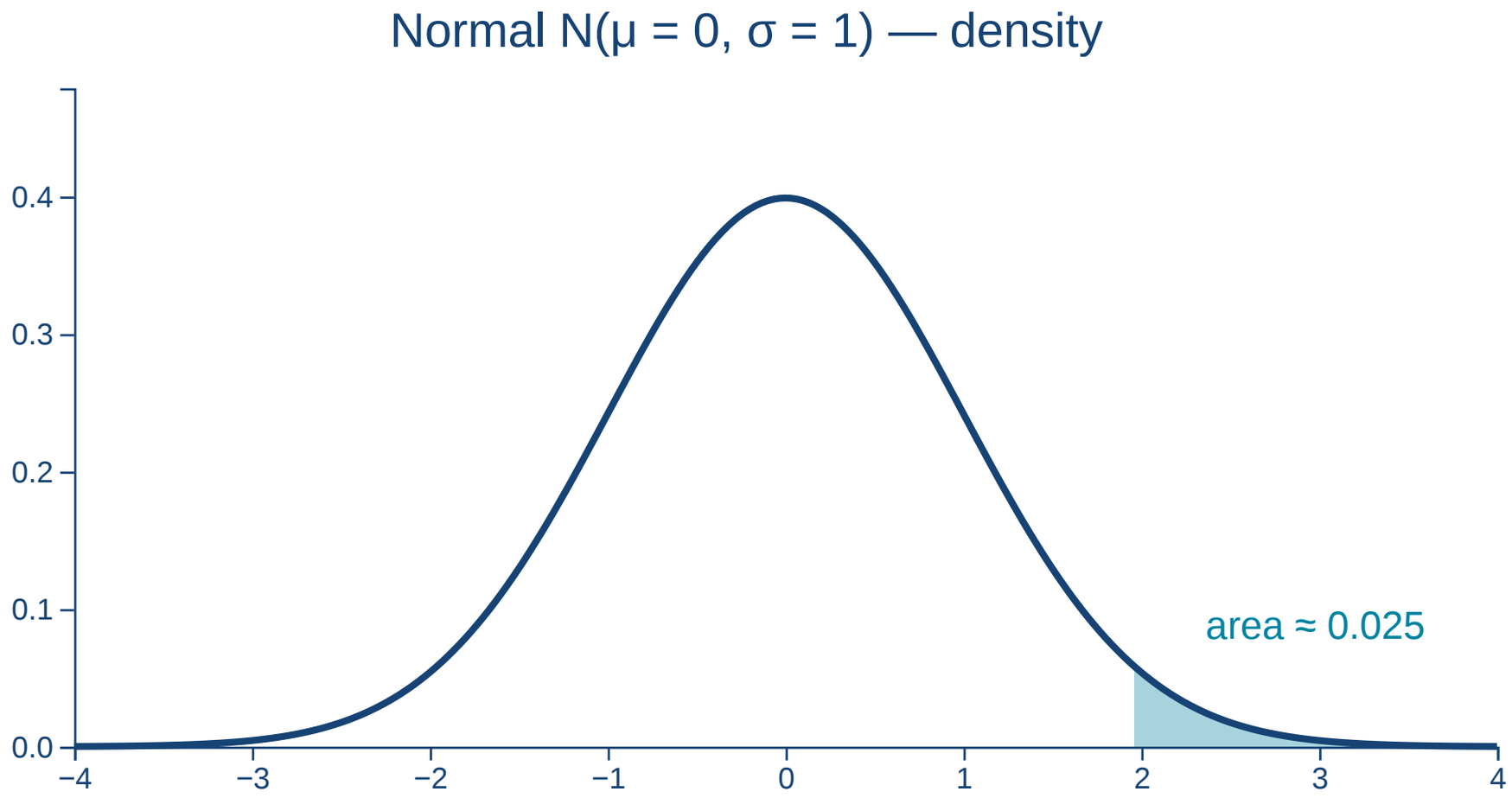
We never prove H_1. We only measure how badly H_0 fails to explain the data.

Test Statistic and p-Value

A **test statistic** $T = T(X_1, \ldots, X_n)$ compresses the data so that its distribution **under** H_0 is known.
The **p-value** is the probability, under H_0, of observing a statistic at least as extreme as the one we got.
Small p-value → the data is surprising under H_0.

> The p-value is **not** the probability that H_0 is true. It is $P(|T| \geq |t_{\text{obs}}| \mid H_0)$ — a tail probability of the statistic — not $P(H_0 \mid \text{data})$.

The Rejection Region



Under H_0 the statistic follows a known distribution. The shaded tail carries probability $\alpha/2$: if T lands there, we reject.

Two Ways to Be Wrong

Type I error — false alarm
Reject H_0 when it is true.
Probability α (we choose it).

Type II error — miss
Fail to reject H_0 when H_1 holds.
Probability β; **power** $= 1 - \beta$.

Lowering α raises β for a fixed design. To improve both: more data, less noise, or a **paired** design that cancels shared noise.

One Sample: z or t?

Known variance — z-test

$$Z = \frac{\overline{X} - \mu_0}{\sigma/\sqrt{n}}$$

Reject (two-sided) if $|Z| > z_{1-\alpha/2}$.

Unknown variance — t-test

$$T = \frac{\overline{X} - \mu_0}{s/\sqrt{n}}$$

s estimated from the data → **heavier tails** (t with $n-1$ d.f.).

$\sigma/\sqrt{n}$ is the **standard error**: the standard deviation of the sample mean $\overline{X}$. $s = \sqrt{s^2}$ is the sample standard deviation. Both statistics measure the distance from μ_0 in standard errors.

Two Samples: Which t?

Situation	Test	Statistic
independent runs, equal variances credible	pooled t	$\frac{\overline{X}_B - \overline{X}_A}{s_p\sqrt{1/n_A + 1/n_B}}$
independent runs, variances may differ	**Welch's t**	$\frac{\overline{X}_B - \overline{X}_A}{\sqrt{s_A^2/n_A + s_B^2/n_B}}$
both models on the **same** seeds or splits	**paired t**	t-test on the differences $d_i = x_{B,i} - x_{A,i}$

> Welch is the default for independent runs. When both models can be run on the same splits, pair them: the shared noise cancels in d_i.

Testing Variances

Single variance (normal model), χ^2 statistic with $n-1$ degrees of freedom.
Is run-to-run variability within specification?

Two variances, $F = s_A^2 / s_B^2$.
If the variances differ, **do not pool —** go back to Welch's t for the means.

Back to the A/B Question — Worked

Ten runs each. A: mean accuracy 80 %, $s_A = 2$. B: mean 82 %, $s_B = 1$.

$$t = \frac{82 - 80}{\sqrt{4/10 + 1/10}} = \frac{2}{0.71} \approx 2.83$$

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> Paired on the same ten seeds, with differences of mean 2 and $s_d = 0.5$: $t = 2/(0.5/\sqrt{10}) \approx 12.6$. Pairing buys far more power than more runs.

The Recommendation

Default

Welch's two-sample t on the mean difference — or paired t if both ran on the same splits

Care about stability

F-test on s_A^2 / s_B^2

Normality dubious

Permutation (randomisation) test on the mean difference

Hypothesis testing compares data to a model through a statistic whose null distribution is known. Known vs. unknown variance changes that distribution (z vs. t); the design changes the test (Welch for independent runs, paired for shared splits).

The Churn Shop: Is the Difference Real?

Lecture 01's dashboard, with counts. Over one year:

	customers	churned	rate
app users	400	40	10%
no app	400	60	15%

H_0: both groups churn at the same rate. Pooled rate $\hat{p} = 100/800 = 0.125$.

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> Chance alone would rarely produce a gap this large: at $\alpha = 0.05$ the difference is **real**. Whether the app *causes* it is a different question — Lecture 03 draws both answers.

Image Credits & References

Sheldon, Ross and others (2002). A first course in probability. *Pearson Education India*.